

ABSTRACT

DESIGN AND IMPLEMENTATION OF ADVANCED SENSING PLATFORM FOR WEARABLE BIOMEDICAL APPLICATIONS DEVELOPMENT

By

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The objective of this thesis is to design and implement a portable, smart body sensor device will allow anyone to obtain important vital (physiological) data anytime and anywhere. These vital data include physiological activities, skin/core body temperature, heart rate, and electrocardiography (ECG). Moreover, the designed portable device could in future leverage smart devices to connect to Cloud framework, which will create a virtual interface with physicians and other authorized individuals to access an individual's health data. This smart body sensor service system could be used as continuous monitoring device that alerts the doctor/center when some critical value is observed. Proposed design could be a paradigm shift for monitoring the critically ill, personnel working in high-risk professions or any individual in the society. By streamlining doctor visits and providing health feeds in real time, this approach could offer a solution to the shortage of family doctors in the US.

DESIGN AND IMPLEMENTATION OF ADVANCED SENSING PLATFORM
FOR WEARABLE BIOMEDICAL APPLICATIONS DEVELOPMENT

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CHAPTER 1

INTRODUCTION

Most people live their life with the simple assumption that every organ and cell in their body is working perfectly 24/7, 365 days a year, unless they feel some specific symptom indicating otherwise. The human body is a scientific marvel and has amazing healing powers that enable it to bounce back from a vast array of illnesses and injuries. Studies show that possessing accurate health data enables us to understand how our body is functioning day by day, and can help in preventing any serious illnesses in the future. Using this continuous monitoring cause-effect data can be more objectively monitored; data can be used for auto-feedback and stress reduction to help the individuals live healthier and reduce healthcare costs by preventing causes of disease. Body sensor network technologies have advanced enough to be used for improving the quality of life, and are considered one of the key research areas in engineering and healthcare application industries, due to their ability to provide one's health data in a matter of seconds, on demand.

Advancements in microelectro-mechanical systems, nanotechnology, and wireless communications have motivated the development of small and low power biomedical sensors and radio equipped modules which are now replacing traditional wired sensor systems. The technology advancements in electronics have reduced the production costs and have made it possible to afford inexpensive sensor devices for ordinary users as well. Together with the mature and also inexpensive wireless technology (RFID, Bluetooth, Wi-Fi, ZigBee, etc.), the costs for enhanced pervasive healthcare systems are within the affordable range of many people.

Being able to identify the context is another benefit achieved with pervasive healthcare systems. Context-awareness enables us to understand the conditions of the people to be monitored constantly and the environments in which these people reside. This contextual information is achieved mostly by sensing systems that incorporate more than one type of sensing capabilities. By fusing the information gathered by several sensors, we may have a more clear understanding of the context. The context information helps in identifying any unusual patterns and enables the making of more precise inferences in any given situation through sensor fusion.

To build the foundation of the portable body sensor networks by using latest state-of-art components, this thesis demonstrates a complete implementation of portable hardware platforms which consists of different blocks such as ECG front end, GSR (galvanic skin response), proximity sensor, temperature sensor, motion processor, Bluetooth, power module and others. All of these components are connected to a single ARM Cortex M3 MCU (LPC1549). The front end of an ECG must be able to deal with extremely weak signals from 0.5 mV to 5.0 mV. The meaningful bandwidth of an ECG signal ranges from 0.5 Hz to 50 Hz. ECG signals may be corrupted by various kinds of noises, hence to reduce these noises a robust hardware architecture is presented in this thesis.

The thesis is organized as follows: In chapter 2 we explain related works especially regarding available hardware platforms and their capabilities focused to body sensors networking. In chapter 3, a detailed description of the proposed architecture is provided; we explain data acquisition methods for the designed hardware and also analyze the acquired data at chapter 4 and finally we conclude the thesis with possible updates at chapter 5.

CHAPTER 2

RELATED WORKS

Overview

One of the main issues related to such body sensor networks or Body Area Networks (BAN) is power consumption, since changing batteries constantly is a burdensome task. Therefore, development of energy efficient Medium Access Control (MAC) protocols and energy efficient sensor devices is critical [1]. The proper use of the wireless communication channels during routing ensures energy efficiency [2]. Moreover, using very low power Pyroelectric Infrared (PIR) sensors [3] enables a simple yet efficient remote monitoring application to be developed. Several studies have investigated the bio-effects caused by the radio frequency transmission of sensor nodes, including thermal and thermal effects and control algorithms to reduce the bio-effects [4]. Another important design constraint for such a system is unobtrusiveness. Yoo et al. [5] have designed an adhesive bandage-type ECG sensor that is wirelessly powered by a health monitoring chest band. Another approach [6] is also trying to provide continuous health and activity monitoring for elderly people who are commonly living alone. Sensor technologies in nanotechnology might allow these sensors to be much smaller, consuming far less power. In some cases it may be possible to simply “spray” these sensors onto people and places allowing the further extension of what information can be gathered in a non-invasive manner [7,8].

Mobility and portability are other considerations for the physical design of such sensor systems, since patients have to wear such BAN devices all the time and mobility reduction is not acceptable. The sensor devices must be designed with an aim

of providing the highest degree of mobility for the patients [9, 10], which necessitates the integration of several network technologies like RFID and near field communication [11]. An example of such a system is a smart shirt which measures ECG and acceleration signals continuously and transmits the physiological ECG data and physical activity data to an IEEE 802.15.4 ad hoc network [12]. Other studies also make use of ZigBee and mobile phones for ECG and blood glucose measurements [13]. In a similar study, a sensor that can be integrated into clothing is designed to measure biochemical changes in sweat which may indicate some health related problems. The study has shown that the sensor has the potential to record real-time variations in sweat during exercise [14]. The real-time availability and reliable communications of a networked body sensor system is another major issue, since the data gathered by this system may be life critical [15, 16]. Finally, security issues must also be considered, since the physiological data of the individual is highly confidential. Progress has been made in designing a physiological, signal-based entity authentication scheme [17].

Existing Approaches

Today, many of biosensors are available in the market to measure different parameters of our body. A list of some of biosensors and their application are shown in Table 1. Body-sensor network (BSN) is a wireless network of wearable computing devices that are used in wearable health-monitoring system (WHMS). Also Scanadu, a Silicon-Valley based company is working on BSNs and its products like Scanadu-Scout is a major attraction in health monitoring systems.

TABLE 1. Different Types of Biosensors

Types of bio-signal	Types of sensor	Description of measured data
Electrocardiogram(ECG)	Skin/Chest electrodes	Electrical activity of the heart
Blood pressure(Systolic and diastolic)	Arm-cuff based monitor	Refers to the force exerted by circulating blood on the walls of blood vessels
Body and/or Skin temperature	Temperature probe or skin patch	Fever/Body Temperature monitoring
Respiration rate	Piezoelectric/piezo-sensitive sensor	Number of movements indicative of inspiration and expiration per unit time
Oxygen saturation	Pulse oximeter	Indicates the oxygenation or the amount of oxygen that is being carried
Heart rate	Pulse oximeter/skin electrodes	Frequency of cardiac cycle
Perspiration or skin conductivity	Galvanic Skin Response	Electrical conductance of the skin is associated with the activity of the sweat glands
Heart sounds	Phonocardiogram	Heart Phonocardiogram pattern record
Blood glucose	Skin electrodes	Blood Glucose Estimation with skin resistance
Electromyogram(EMG)	Skin electrodes	Electrical activity of the skeletal muscles
Body movements	Accelerometer	Measurement of acceleration forces in 3D space

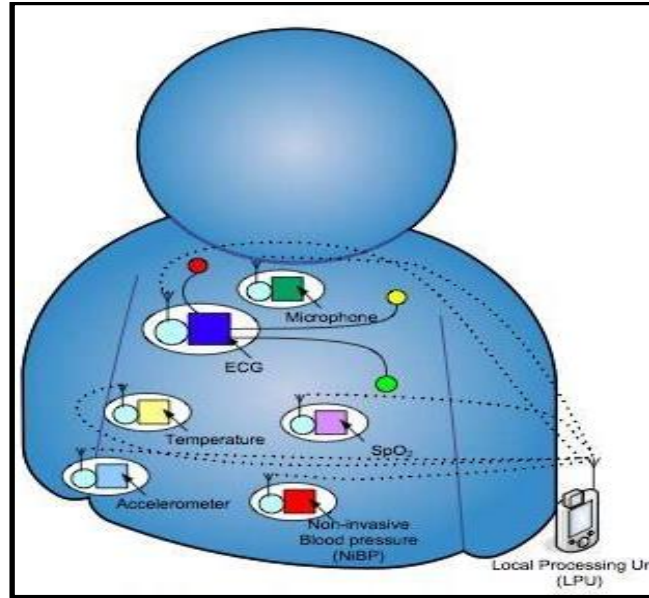


FIGURE 1. Wearable health monitoring system using bio-sensors.

Wireless sensor networks (WSNs) consist of low-power embedded computing devices such as motes which are called platforms. Fig. 1 above shows how a person's body is monitored by making Body Sensor Network using different biosensors. This type of system is beneficial to the medical field as any ill person does not need to go to a doctor's clinic for his/her body check-up. He/she can have the report of his/her body by just sitting at home or any other place [18, 19].

Here we will discuss some of the existing platforms (SHIMMER, Mercury, TelosB, Egs, DexterNet, SPINE, ATOM) in Body Sensor Networks. Moreover a comparison table is made of all the discussed platforms.

SHIMMER and Its Implementation In Software Module-BioMOBIUS

The SHIMMER platform is a flexible sensing platform for patient monitoring and diagnostic which consists of specific power management hardware, open software with advantages over other research platform. In this platform modification can be

done by changing software device configuration and by physical means which requires less complex software and hardware support. This sensing system is improved by placing transducer, signal conditioning and other subsystem on single daughter board which connects to baseboard thus providing enhanced communication and computational capabilities. It deploys connectors for expansion which supports auxiliary daughter boards [20].

There are two types of add-on boards, one is standard and other is motherboard. A standard board consists of two types of sensors, humidity and temperature, and motherboard includes various external communication for GPRS, WLAN and options for data storage through SD card. This platform is extensively used for Gait analysis, alertness training platform and in home gait velocity measurement.

SHIMMER platform consist of baseboard which contributes to sensors computations, communication, data storage and daughterboard connection capabilities. The core features are extended via a range of daughter boards which gives different physiological, kinematics and ambient sensing technology [21].

Core computation. As it is illustrated in Fig 2. the SHIMMER uses Texas Instruments MSP430 which uses wireless sensors. The main advantage of this MCU is its extreme low power feature in idle period. In SHIMMER, micro SD card socket is connected via SPI. In synchronous mode, external system can be connected via USART using three or four pin. The MCU has 8 ADC channel using 12 bit A/D conversion. In SHIMMER, external ports can be used for data input from XYZ accelerometer, external expansion connector and internal expansion connector. When ADC is not used it is disabled to support low power capabilities.

Communications. Platform uses Chipcon CC2420 radio transceiver which uses IEEE 802.15.4 wireless communication. For low power consumption radio may be turned off by controller. Radio is controlled by an SPI. The Bluetooth module is connected to the controller Via USART1.

Micro SD storage. A SHIMMER base board has Micro SD card socket which supports storage up to 2 Gigabytes. This supports storage of non-streaming application and secures data while mobile, changing batteries and during network outage.

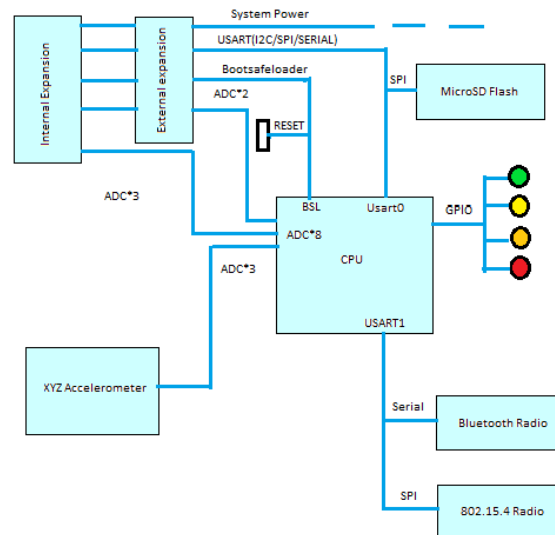


FIGURE 2. SHIMMER core computation.

Power Management. Reducing intervals of communication between radio and micro SD card can extend the battery life but not an optimal solution in case of high sample rate data capture required. The micro SD card, accelerometer, 802.15.4 radio and Bluetooth are powered off by firmware when not in use.

Shimmer Expansion. SHIMMER baseboard has external and internal connectors to expand platform. At the internal expansion, user can connect daughterboard such as kinematics, Electrocardiogram (ECG) and galvanic skin response to baseboard. At external expansion, user can connect the board to external programming port.

SHIMMER Sensing Capabilities

Kinematic sensing. The accelerometer is linked to MSP430 CPU using three channels of ADC. The daughterboard of gyroscope uses pair of MEMs gyroscopes. The flex segment rotates around the module for proper orientation of sensor for XYZ measurements. The magnetometer daughterboard uses gyroscope and magneto-resistive gyroscope sensor for low field magnetic sensing.

Physiological sensing. The ECG daughterboard contains low power CMOS operational amplifier. The lead input contains weak pull-up for the detection of floating electrodes. The gain of an amplifier is 175 with frequency response of 0.05 to 150 Hz.

Ambient sensing. The PIR motion daughterboard uses a Panasonic NaPiOn passive infrared motion sensor. It is low power and extremely compact. The ADL daughterboard has light and temperature sensor.

External hardware integration. An analog expansion module combines two analog signals to be combined into SHIMMER. AnEx is simple external board for experiment and prototyping. It uses Maxim MAX768EEE charge pump for boosting SHIMMER 3V supply to +5 volt. PRIMMER is daughterboard which extends the SHIMMER's internal connector. The goal of PRIMMER is to integrate commercially available and existing physiological sensor circuitry integration.

Shimmer firmware. SHIMMER firmware runs on tinyOS. This programming environment helps different low power devices with small amount of memory and wireless communication capabilities. It manages radio communication, task scheduling time, I/O processing, with very little size of footprint. Current firmware functionality contains FAT system, Micro SD flash storage, peripheral configuration and control and power supply monitoring. Sensor data can be streamed at rate to 500Hz in real time to BioMOBIUS application via Bluetooth radio. To enhance reliable communication a new packet format is implemented which supports reliability and different sensor data through cyclic redundancy and packet sequence number technique.

This firmware is used with BioMOBIUS and frame format supports data reliability. The packet contains binary data without restriction and it can lead to issue hex values appearing in packet header or payload is interpreted as control characters like End of frame, Beginning of frame. To avoid this byte stuffing is used which is based on HDLC protocol. The File allocation table (FAT) file system is most used and it provides a high level API which supports manipulating files on SD card.

Power management using firmware. Battery Optimization for SHIMMER has been implemented for firmware. Primary optimization is implemented in firmware with goal of lowering the duty cycle. Other implementation is low battery indication which prevents the corrupted data to be sent to BioMOBIUS software application. The node has also capability to detect low voltage condition. The accelerometer needs battery voltage to be constant in order to send correct data. When battery voltage falls below +3V SHIMMER firmware halts data collection.

SHIMMER Integration with BioMOBIUS Software

It is fully supported with BioMOBIUS high level application. It is very convenient for application developer to acquire data from SHIMMER node, process and present it through GUI and finally store data to file or database. Communication is done by SHIMMER sensor to BioMOBIUS via SHIMMER block. This block is general block which Transmits and receives data from SHIMMER sensor and generates output data on its output pin. The data that presents on output pin can be raw if data is not processed.

BioMOBIUS is nothing but the software environment for implementation of bio-medical data sets. It comprises of GDE Eyes Web which creates a patch and also a GUI tool to control and enable user to interface the patch. Fig. 3 shows the block diagram of BioMOBIUS.

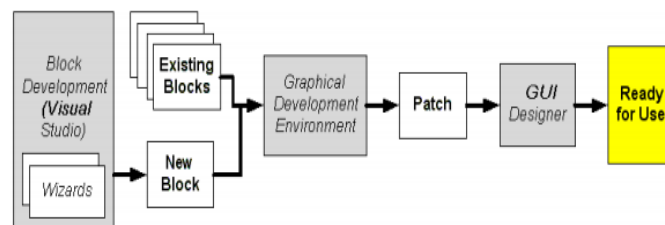


FIGURE 3. Taxonomy of BioMOBIUS.

Now to implement SHIMMER platform in BioMOBIUS it required to follow four steps as follows:

Connect SHIMMER to BioMOBIUS using Bluetooth

Create Path using EyesWeb

Create UI application to link to patch

Running the application in run-time environment. As shown in Fig 4. a patch is made first of all using EyesWeb GDE inside BioMOBIUS. This is made to control SHIMMER, and to acquire data from SHIMMER.

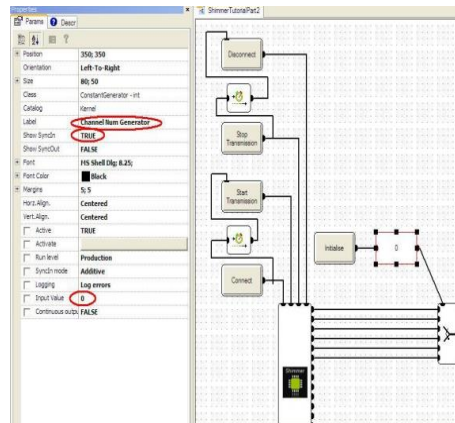


FIGURE 4. Creating a patch using EyesWeb GDE.

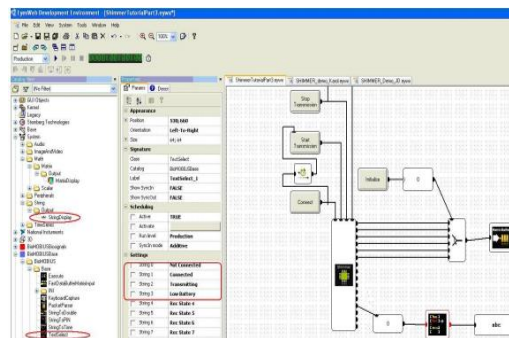


FIGURE 5. Connection of SHIMMER ADC and other pins inside EyesWeb patch.

Finally a UI Designer designs a UI in GUI of BioMOBIUS and one can have control over SHIMMER and can also get the output of raw data along with the SHIMMER status as shown in Figure 6.

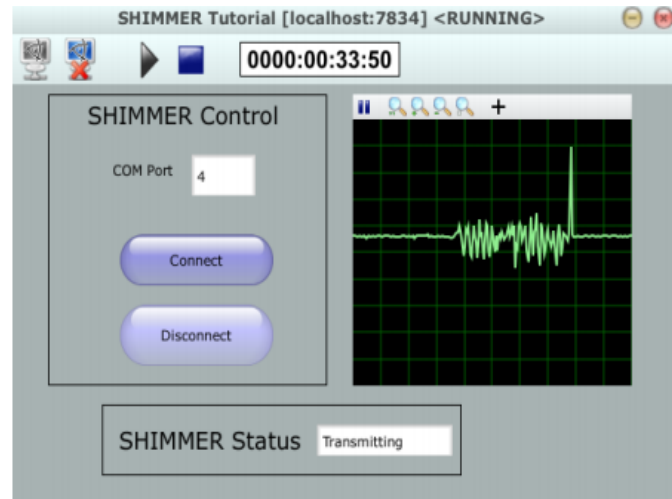


FIGURE 6. GUI of SHIMMER in BioMOBIUS.

So till creating a patch in EyesWeb and connection of pins in it works as a back-end process and after UI designer makes its corresponding UI that works in front-end of the whole process. In this way, SHIMMER platform is integrated with its corresponding software module called BioMOBIUS.

Mercury Platform and Its Architecture

Mercury is wireless sensor platform used for long term data collection on patients suffering from Parkinson's disease, neuromotor disorder, epilepsy and stroke. Individual nodes compute the high level properties from raw signal and base station collects data and tunes the different parameters based on different conditions. It is

designed to overcome long battery life high data fidelity challenges. It provides high level programming interface which allows the researcher to quickly develop different policies for data driving and tuning lifetime of sensor. Wireless data have major challenge in energy constraint and fluctuation with radio link. Continuous transmission of sensor data would rapidly deplete the sensor node battery. Therefore some data reduction is needed. Mercury provides feature extraction from raw sensor data performed on sensor node. Generally these features are smaller than original so bandwidth and energy are conserved [22].

Hardware platform and energy profile. Gyroscope utilizes large amount of energy. Power consumption is reduced by disabling gyroscope sampling when node is not in movement. It is not worth computing more computational feature like FFT on sensor node. Node is continuously sampling gyroscope and accelerometer data to maintain time sequence but does not transfer data to base station. Efficiently duty cycling gyroscope and flash logging reduce notable energy consumption. A typical mercury network consists of multiple sensor nodes that can be wearable by patient and a base station which can be laptop with 802.15.4 transceiver. Software is divided into two parts that run on base station and sensor node. Sensor nodes receive and store data, catch features, and respond to query from base station for data download and changing the sample rate. The main application specific logic exists in driver running on base station. Mercury deliver simple API to driver for controlling operation of sensor node and data retrieving. The mercury driver implement policies for storage, tuning data sampling.

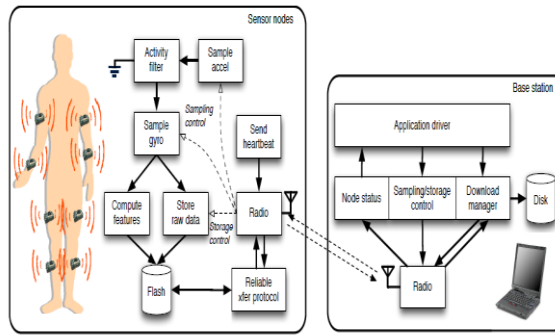


FIGURE 7. MERCURY Architecture.

Sensor node software. Sensor node allocate narrow interface which allow the base station driver to tune storage mode and sampling rate. This eliminate the need for clinical to program sensor node software. Pixie operating system is used to implement sensor node software.

Sampling and activity filter. In normal condition Mercury node samples its different sensors in sample blocks; and each sample block hold a piece of raw sensor data. For storage and processing; sample blocks are passed to downstream modules. Base station control set of active channel and sampling rate in mercury node via interface provided by node. Mercury contain activity filter that conserve energy when sensor in not active. Accelerometer determine whether sensor is not moving if it is not moving then filter drop the data and disable gyroscope. When sensor begin to some movement activity filter begin in active period. In this period all samples are sampled by current sampling rate.

Feature extraction. Mercury provide feature extraction which are efficient to compute on sensor node. Some application may need different (custom) algorithm for calculation. Based on research study; mercury provides five standard feature algorithms: mean, maximum peak-to –peak amplitude, RMS, peak velocity, and RMS

of jerk time series. These mercury features can be computed for six channels of gyroscope and accelerometer data.

Heartbeats. Sensor nodes repeatedly transmit heartbeat packets to base station which contains essential information for data retrieval schedule. The base station includes the feature blocks and number of samples stored on a node. Heartbeats are transmitted every 60 second interval in inactive stage or when node stores new feature block. Heartbeats contain radio link quality to base station and node's energy consumption.

Reliable transfer protocol. Mercury provides simple reliable end to end transfer protocol. The base station transmits a request which contains sensor node ID, features or raw data, and range of block address which to retrieve. To conserve energy for sensor node use low power listening MAC that duty cycle radio for incoming packet.

Application driver. The core of system is application driver that runs on base station and manage the sensor network operation like data acquisition, storage, sampling. Driver is important in implementing policies that are that control data quality, sensor lifetime. Mercury contains API that permit driver to configure for data download, sampling and storage modes. Different operations query the state of sensor node on base of updated heartbeat received. As an example; energy consumed (n) command get the energy consumed data on node (n) with last updated heartbeat received. Downloads are done using download sample and download feature operation.

TelosB And Egs Platforms

TelosB Mote. MEMSIC's TelosB Mote TPR2420 is an open-source platform designed to support lab studies and research community. The TelosB Mote TPR2420

contains all the necessary parts for labs and research into a single platform. It has USB programming capability and runs TinyOS 1.1.11 or higher. An IEEE 802.15.4 compliant RF transceiver with integrated antenna is built on-board. It uses a low-power MCU, 8MHz TI MSP430 microcontroller with 10kB RAM and 1MB external flash for data logging. TPR2420 offers an optional sensor suite including integrated light, temperature and humidity sensor.

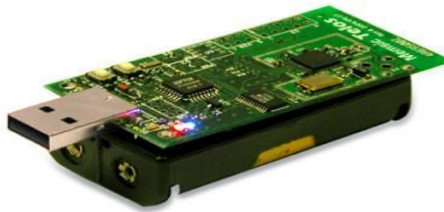
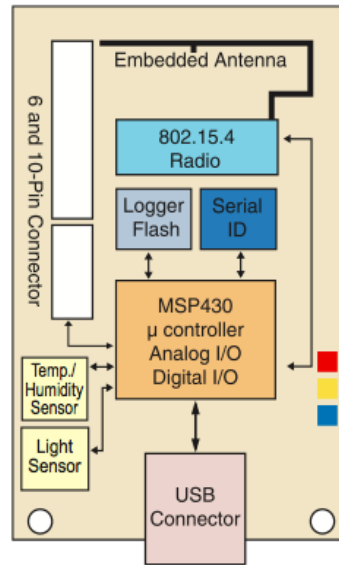


FIGURE 8. TelosB Mote.

The TelosB platform was developed by UC Berkeley. This platform is used for long battery life because it has low power consumption and fast wakeup from sleep state. It is powered by two AA batteries or plugged into the USB port for programming or communication.

The TPR2420 is also powered by host computer. TPR2420 provides interface with additional devices. By using the two expansion connectors and onboard jumpers, analog sensors, digital peripherals and LCD displays can be controlled. The TPR2420 is compatible with the open-source TinyOS distribution developed by UC Berkeley. TinyOS is a small, open-source software operating system which supports self-

configuring, large scale sensor networks. The software development tools for source code are publicly available [23].



TPR2420CA Block Diagram

FIGURE 9. TelosB internal block diagram.

Egs Mote. Egs Mote platform is designed to increase computational power and maintain low energy consumption. The Egs Mote platform is based on the Cortex M3 processor, a 32-bit ARM processor based on the ARMv7-M architecture. Egs Mote platform uses the Atmel SAM3U2C with ARM Cortex M3 core to increase computational power but still keeps low energy consumption.

Atmel SAM3U2C is combined with a Bluetooth and a Zigbee radio for wireless communications in a single platform. Egs platform also provides an on-board micro SD card slot, a 2Gb NAND flash, and a 2.4 inch touch screen LCD. Egs Mote platform is used for medical sensing applications, so it is connected to a pulse

oximetry sensor board, and a fingertip pulse oximetry sensor by a UART port, and a DB9 connector.

Hardware platform. The main design goals of Egs platform are energy efficiency and computational power. It can provides wireless sensor network developers with a hardware platform that has faster processing speed, more memory size and data storage. In addition, it must be applicable to long-term, unattended deployments with low power consumption and high energy efficiency.



FIGURE 10. Egs Mote.

To achieve those goals, Egs Mote platform relies on Atmel's SAM3U2C microcontroller that uses the 32-bit ARM Cortex-M3 processor as its core, has a SRAM size of 36 KB, flash size of 128KB and it provides control for various peripheral controllers, and a memory protection unit (MPU). The SAM3U2C is more

powerful than MSP430, and it can run up to 96 MHz with small required current, 48 mA in active mode.

The SAM3U2C allow the integration of various external components. The Egs platform has large amounts of data storage a 2 Gb NAND flash and an external micro SD card slot. In addition, the Egs platform supplies a TI CC2520 802.15.4 radio and Mitsumi's class 2 Bluetooth module both integrated with chip antennas for wireless communication. Especially, a 2.4 inch LCD touch screen with a resolution of 240×320 pixels that allows interface with user covers the front side of Egs platform. The device is powered with a mini-USB connection or a rechargeable battery for mobile operations. The new version of Egs platform can connect with Nellcor pulse oximetry sensor board and fingertip sensors by UART pins and DB9 pins connectors.

The Egs platform supports to TinyOS 2.x and the implementation supports various peripheral controllers: USART, UART, ADC, I2C, SPI, and DMA controllers. These peripheral controllers provide drivers for the external components.

DexterNet Platform and SPINE Framework

DexterNet hardware and applications. DexterNet platform is an open platform for heterogeneous body sensor networks. DexterNet is designed to bear a signal processing library, SPINE. This Signal Processing in Node Environment, called SPINE, is an open-source library that can support many different types of body sensors. Therefore, the higher level applications on DexterNet platform can control various body sensors. DexterNet can be used to collect real- time signal wirelessly and manage many different body sensors depending on different application needs. DexterNet platform uses three-layer BSN architecture to monitor multiple subjects in both indoor and outdoor environment. The three-layer BSN architecture includes a

body sensor layer (BSL), a personal network layer (PNL), and a global network layer (GNL).

The following Fig 11. Shows the three-layer hierarchy of DexterNet. The BSL contains many commercial and custom-built body sensors that connect with multiple sensor network nodes to form wearable sensor nodes. The PNL is a wireless mobile station that communicates with multiple body sensors wirelessly. The SPINE library supports on BSL and PNL to manage the data processing, collection and transmission of the data

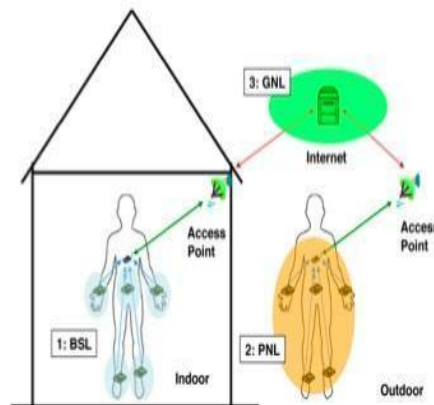


FIGURE 11. Three layer hierarchy of DexterNet.

At GNL, a complete pathway for sensing, processing, wireless communication, and data fusion are ready for higher-level application. With DexterNet platform, various applications can be developed with different body sensors and sensor network systems.

At the BSL and PNL, the open-source SPINE framework supports many different physical components such as TelosB mote, MICAz mote, and SHIMMER mote. This flexibility of SPINE and its separation in code of sensing, processing, and communication make SPINE portable across TinyOS and extendable. It uses sensor drivers to support new sensors. With SPINE library and three-layer architecture, DexterNet becomes a high competitive framework for healthcare, military, and consumer electronic applications. For example, human activity recognition application can be implemented at the PNL level to collect the data measured at different location of the body. Applications can be built on top of the SPINE API, so they can configure the appropriate sensors and signal processing depend on their goals and requirements. These applications do not require depending on any specific sensor drivers. Moreover, developers can work on application-level software and SPINE software at the same time without any problem in traditional system.

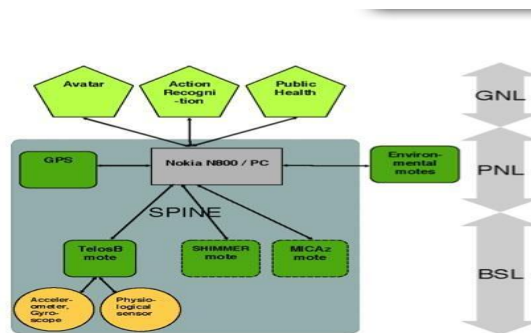


FIGURE 12. Layers and SPINE Framework.

Hardware of DexterNet. A variety of sensors and motes can be integrated into the system. Currently, DexterNet platform supports three types of sensors. Those are

physiological sensors, motion sensors, and GPS (Global Positioning System) sensor. In addition, many motes such as SHIMMER mote and MICAz mote has many onboard sensors like accelerometer or available as add ons, including sensors such as humidity, ambient light, barometric pressure, GPS, magnetometer, sound, etc.



FIGURE 13. DexterNet Hardware Mote.

Recently, a motion sensor node with a triaxial accelerometer and a biaxial gyroscope is designed and added on a TelosB mote for wireless network communication and on-node signal processing. The real-time data is collected and transfer wirelessly by using 802.15.4 wireless protocol with sample rates up to 64 Hz. The TelosB mote has the onboard microcontroller MSP430F1611, so simple signal processing tasks can be done in real-time. The MSP430 microcontroller runs at 8 MHz clock frequency with 10 kB of internal RAM and converses analog data signal to 12 bit digital signal. The 600mAh battery can supply enough power for 20 hour operation with continuous measurement and raw data transmission wirelessly.

Applications. DexterNet can be used as a body sensor network platform in many applications to monitor human actions or activities. For example, Avatar is an application that uses a network of motion sensors to reassemble and envision a person's full-body motion in real time. The application can remotely monitor and

assess the well-being of elderly people living alone, or record and visualize the movements of patients.

As a conclusion of this chapter TABLE 2. is the comparison of available platforms has been added.

TABLE 2. Comparison of Available Platform

Sensor Node Name	Microcontroller	Transceiver	Program and Data Memory	External Memory	Programming	Remarks
COOKIES	ADUC841, MSP430	ETRX2 TELEGESIS, ZigBit 868/915	4Kbytes + 62Kbytes	4Mbit	C	Platform with hardware reconfigurability(Spartan 3 FPGA Actel Igloo)
BEAN	MSP430F169	CC1000 (300-1000MHz) with 78.6kbit/sec.		4Mbit	C	YATOS Support
BTnode	Atmel ATmega 128L (8Mhz @ 8 MIPS)	Cjipcon CC1000 (433-915MHz)and Bluetooth 2.4GHz	64+180KRAM	128KB Flash ROM, 4KB EEPROM	C and nesC Programming	BTnut and TinyOS Support
Egs	ARM Cortex M3	CC2520, Mitsumi's Class 2 Bluetooth	-----	2Gbit	-----	TinyOS Support
IMote 2.0	Marvell PXA 271 ARM 11-400 MHz	TI CC2420 802.15.4/ZigBee compliant	32MB SRAM	32MB FLASH	-----	Microsoft .Net Micro, Linux, TinyOS Support
Iris	ATmega 1281	Atmel AT86RF230 802.15.4/ZigBee compliant	8KB RAM	128KB FLASH	nesC	Mote Runner, TinyOS and MoteWorks Support
MicaZ	ATMEGA 128	TI CC2420/802.15.4/XigBee	4KB RAM	128KB FLASH	nesC	TinyOS, SOS, Mantis OS and Nano-RK Support
Rene	ATMEL 8535	916MHz 10kbit/s	512Bytes RAM	8KB FLASH	-----	TinyOS Support
Sense Node	MSP430F1611	Chipcon CC2420	10KB RAM	48KB FLASH	C and nesC	GenOS and TinyOS Support
Shimmer	MSP430F1611	TI CC2420/802.15.4 compliant	48KB FLASH 10KB RAM	2GB MicroSD card	C and nesC	TinyOS support and full range of expansion modules
TelosB	TI MSP430	250kbit/s 2.4GHz IEEE 802.15.4	10KB RAM	48KB FLASH	-----	Contiki, TinyOS, SOS and Mantis OS Support
Tinynode	TIMSP430	Semtech SX1211	8KB RAM	512 KB FLASH	C	TinyOS Support

VEmesh	TI MSP430	Semtech SX1211/1231 , TI TRF6903	512B RAM	8KB FLASH	Over-the-air programming	FHSS, Interface to MODBUS, DALI, RS 232/485, TCP/IP
Ubimote 1	TI's CC2430 SOC based on 8051 core	TI's CC2430	8KB RAM	128KB FLASH ROM	C	TI's ZStack and TinyOS Support
FireFly	Atmel ATmega 1281	Chipcon CC2420	8KB RAM	128KB FLASH ROM, 8KB EEPROM	C	Nano-RK RTOS Support
T-Mote Sky	TI MSP430	250kbit/s 2.4GHz IEEE 802.15.4 Chipcon Wireless	10 KB RAM	48 KB FLASH	-----	Contiki, TinyOS, SOS and Mantis OS Support

CHAPTER 3

PROPOSED SOLUTION

Hardware Architecture

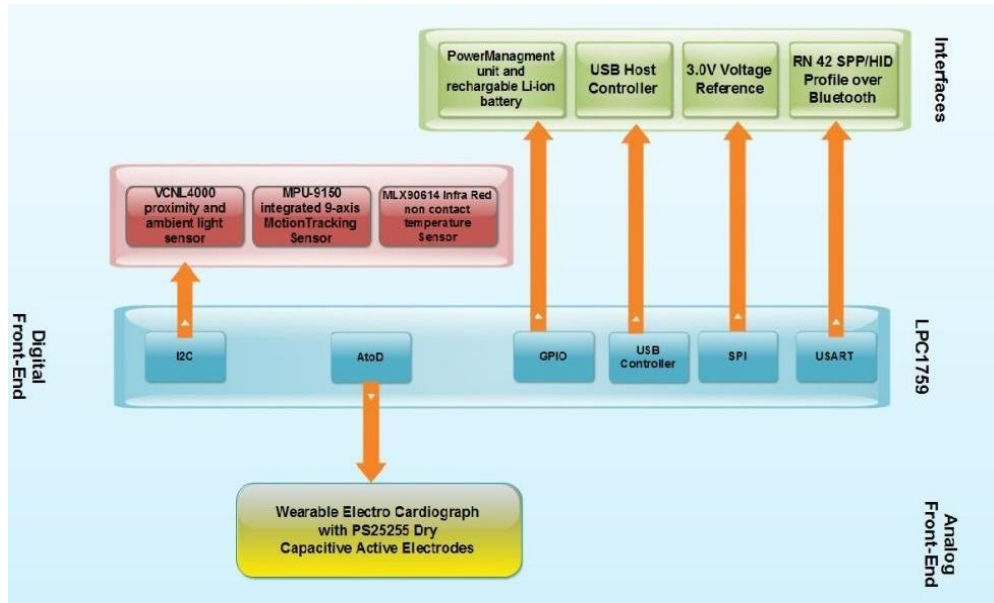


FIGURE 14. Hardware system diagram.

Fig 14. depicts the hardware architecture of the proposed platform. One of the core contributions of this research is to design a power efficient, self-powered, and compact wearable device that can be capable of sensing heart rate, skin/core body temperature, electrocardiography (ECG), proximity and ambient light level of environment. Moreover, it will be able to communicate with the handheld devices (smart phone) using Bluetooth to store the data in a cloud infrastructure.

Firmware Architecture

As shown in Fig 15. The firmware for the developed platform is mainly based on two open source software stacks: Cortex Microcontroller Software Interface Standard (CMSIS) abstraction layer and MBED open source environment. Development of software is a major cost factor in the embedded industry. By standardizing the software interfaces across all Cortex-M silicon vendor products, CMSIS is supporting development of new embedded application or migration of existing software to a new device in a cost effective manner. The CMSIS-CORE defines APIs for a complete system. With the benefits of portable hardware and well-tailored software abstraction, the main focus will be to develop the feasible abnormality diagnosis based on the vital signs observation from our sensors [24,25].

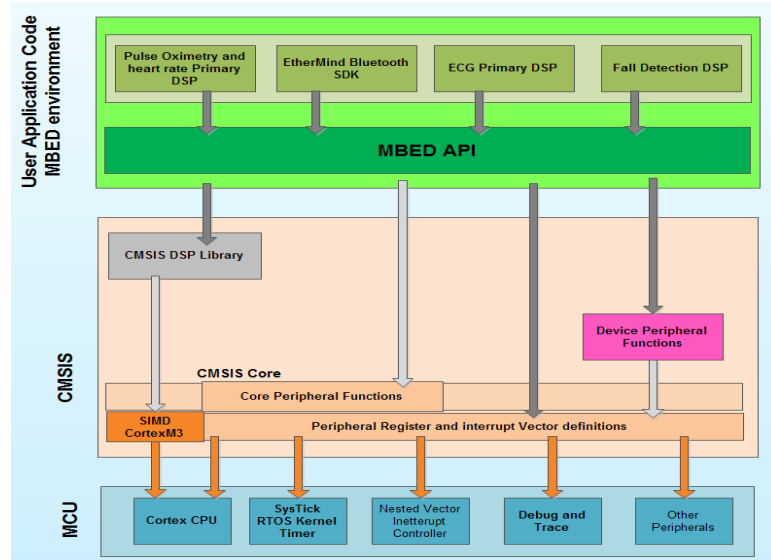


FIGURE 15. Purposed firmware architecture.

Digital Hardware Design Details and Implementation

Power supply management unit

To be able to run the device independently from external power sources for a reasonable period of time, 3.7 volts, 240mAh rechargeable lithium-ion battery has been chosen which is chargeable by a USB source. As illustrated in Fig 16, the MCP73831 chip has been selected for battery charge management.

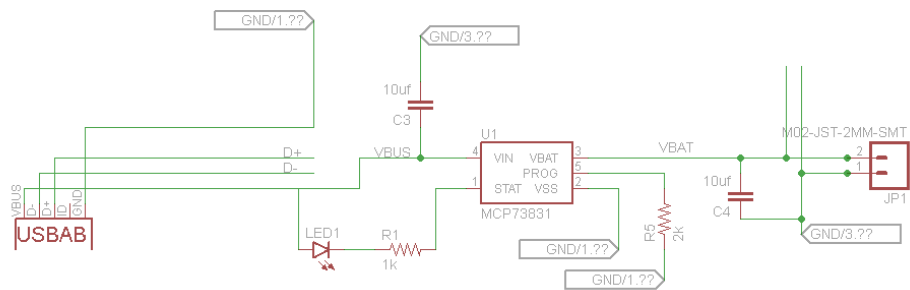


FIGURE 16. Li-Po Battery charger design.

The MCP73831 devices are highly advanced linear charge management controllers for use in space limited, cost-sensitive applications. The MCP73831/2 are available in an 8-Lead, 2 mm x 3 mm DFN package or a 5-Lead, SOT-23 package. Along with their small physical size, the low number of external components requirement makes the MCP73831/2 ideally suited for portable applications. For applications charging from a USB port, the MCP73831/2 adheres to all the specifications governing the USB power bus. The MCP73831/2 employs a constant-current and constant voltage charge algorithm with selectable preconditioning and charge termination [26].

The constant voltage regulation is fixed with four available options: 4.20V, 4.35V, 4.40V or 4.50V, to accommodate the emerging battery charging requirements. The constant current value is set with one external resistor. The MCP73831/2 devices limit the charge current based on die temperature during high power or high ambient conditions. This thermal regulation optimizes the charge cycle time while maintaining device reliability. Fig 17 depicts the operational flow algorithm from charge initiation to completion and automatic recharge [26].

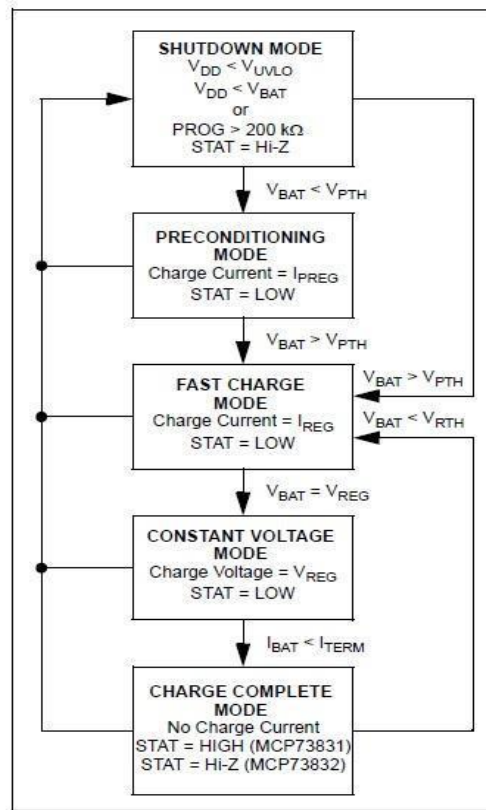


FIGURE 17. MCP73831 charging algorithm.

The VDD is the supply input for the MCP73831/2. The MCP73831/2 automatically enters a Power-Down mode if the voltage on the VDD input falls below the UVLO voltage (VSTOP). This feature prevents draining the battery pack when the VDD supply is not present. Fast charge current regulation can be scaled by placing a programming resistor (RPROG) from the PROG input to VSS.

The program resistor and the charge current are calculated using the Equation 1:

$$I_{reg} = 1000v / R_{PROG} \quad (\text{Equation 1})$$

Where:

$$R_{PROG} = KOhms$$

$$I_{reg} = milliampere$$

The preconditioning trickle charge current and the charge termination current are ratio-metric to the fast charge current based on the selected device options. In Fig 16, in order to achieve 500mAh peak charge current, the Rprog resistor has been set to 2 kilo ohm. Since the whole system (in peak current consumption mode) drains less than 100mA, 500mA is more than enough to achieve convenient charging time for 270mA battery.

The charge status output of the MCP73831 has three different states: High (H), Low (L), and High-Impedance (Hi-Z). The charge status output of the MCP73832 is open-drain, and, as such, has two different states: Low (L), and High-Impedance (Hi-Z). The charge status can be used to illuminate 1, 2, or tri-color LEDs. Table 3 illustrates the state of the status output during the charge cycle [26].

Since the Li-Po battery voltage swipes between 2.5 volts in fully charge state to 4.2 volts out of charge state, TPS61200 low input voltage synchronous boost converter from TEXAS INSTRUMENTS has been chosen to provide steady 3.3 volts

TABLE 3. Charge Status Output in Different Charging Cycle States

Charge Cycle State	STAT1	
	MCP73831	MCP73832
Shutdown	Hi-Z	Hi-Z
No Battery Present	Hi-Z	Hi-Z
Preconditioning	L	L
Constant-Current Fast Charge	L	L
Constant Voltage	L	L
Charge Complete-Standby	H	Hi-Z

For digital and analog blocks while battery voltage swipes during its usage cycle.

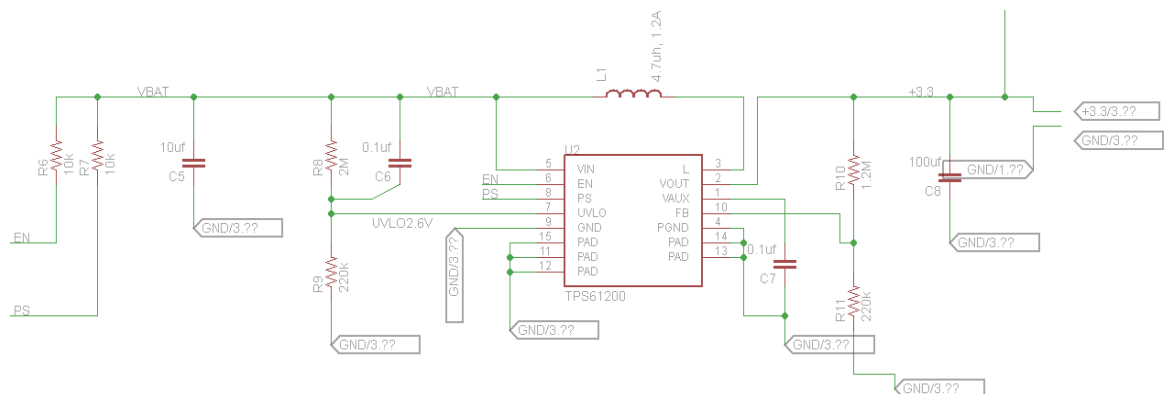


FIGURE 18. Boost converter schematic.

The TPS61200 is illustrated in Fig 18, which provides a power supply solution for products powered by either a single-cell, two-cell, or three-cell alkaline, NiCd or NiMH, or one-cell Li-Ion or Li-polymer battery. The TPS61200 is also used in fuel cell or solar cell powered devices where the capability of handling low input voltages is essential.

Output current depends on the input to output voltage ratio. The device provides output currents up to 600 mA at a 5-V output while using a single-cell Li-Ion or Li-Polymer battery, and discharge it down to 2.5 V. The boost converter is based on a fixed frequency, pulse-width-modulation (PWM) controller using synchronous rectification to obtain maximum efficiency. At low load currents, the converter enters the power save mode to maintain a high efficiency over a wide load current range. The power save mode can be disabled, forcing the converter to operate at a fixed switching frequency. The maximum average input current is limited to 1500 mA. The output voltage can be programmed by an external resistor divider, or is fixed internally on the chip. The converter can be disabled to minimize battery drain. During shutdown, the load is completely disconnected from the battery. The device is packaged in a 10-pin QFN PowerPAD™ package (DRC) measuring 3 mm x 3 mm [27].

The controlling circuit of the device is based on an average current mode topology. The average inductor current is regulated by a fast current regulator loop which is controlled by a voltage control loop. The controller also uses input and output voltage feed-forward. Input and output voltage changes are monitored and can immediately change the duty cycle in the modulator to achieve a fast response to those errors. The voltage error amplifier gets its feedback input from the FB pin. For adjustable output voltages, a resistive voltage divider must be connected to that pin. For fixed output voltages, FB must be connected to the output voltage to directly sense the voltage. Fixed output voltage versions use a trimmed internal resistive divider. The feedback voltage is compared with the internal reference voltage to generate a stable and accurate output voltage. The controller circuit also senses the average input current as well as the peak input current. Thus, the maximum input

power is controlled as well as the maximum peak current to achieve a safe and stable operation under all possible conditions. To protect the device from overheating, an internal temperature sensor is implemented. The device uses three internal N-channel MOSFETs to maintain synchronous power conversion at all possible operating conditions. This enables the device to keep high efficiency over a wide input voltage and output power range.

To avoid ground shift problems due to the high currents in the switches, two separate ground pins GND and PGND are used. The reference for all control functions is the GND pin. The power switches are connected to PGND. Both grounds must be connected on the PCB at only one point ideally close to the GND pin. Due to the 3-switch topology, the load is always disconnected from the input during shutdown of the converter.

A boost converter only regulates output voltages which are higher than the input voltage. This device operates differently. For example, it is able to regulate 3 V at the output with two fresh alkaline cells at the input having a total cell voltage of 3.2 V. Another example is powering white LEDs with a voltage of 3.6 V from a fully charged Li-Ion cell with an output voltage of 4.2 V. To control these applications properly, a down conversion mode is implemented. If the input voltage reaches or exceeds the output voltage, the converter automatically changes to a down conversion mode. In this mode, the control circuit changes the behavior of the two rectifying switches. While continuing switching, it sets the voltage drop across the rectifying switches as high as needed to regulate the output voltage. This must be taken into account for thermal consideration.

The Power Save (PS) pin can be used to select different operation modes. To enable Power Save mode the PS pin must be set low. Power Save mode is used to

improve efficiency at light load. If Power Save mode is enabled, the converter stops operating if the average inductor current goes below 300 mA and the output voltage are at or above its nominal value. If the output voltage goes below its nominal value, the device ramps up the output voltage again by starting operation using a programmed average inductor current higher than required by the current load condition. Operation can last for one or several pulses. The converter stops operating once the conditions for stopping operation are met again. The Power Save mode can be disabled by programming high at the PS pin. In Down Conversion mode, Power Save mode is always enabled and the device cannot be forced into fixed frequency operation at light loads. The PS input supports standard logic thresholds. Since In our case this pin has been set to high and has been disabled to avoid any latency and better response time of the switching power supply.

The device is put into operation when EN is set high. It is put into a Shutdown mode when EN is set to GND. In Shutdown mode, the regulator stops switching, all internal control circuitry including the low-battery comparator is switched off, and the load is disconnected from the input. This also means that the output voltage can drop below the input voltage during shutdown. During start-up of the converter, the duty cycle and the peak current are limited in order to avoid high peak currents drawn from the battery. In our case this pin is assigned to high to keep power supply ON all the time. After being enabled, the device starts operating. At first it keeps the main output VOUT disconnected, and charges the capacitor at VAUX. If the capacitor at VAUX is charged to about 2.5 V, the device switches to normal operation. This means VOUT is turned on and the capacitor at VOUT is charged while the load connected to the device is supplied. To ramp up the output voltage in a controlled way, the average current limit is set to 400 mA and rise proportional to the increase of the output

voltage. At an output voltage of about 1.2 V, the current limit is at its nominal value. If the output voltage does not increase, the current limit will not increase. There is no timer implemented. Thus the output voltage overshoot at startup, as well as the inrush current, is kept at a minimum. The device ramps up the output voltage in a controlled manner even if a large capacitor is connected at the output. When the output voltage does not increase above 1.2 V, the device assumes a short-circuit at the output, and keeps the current limit low to protect itself and the application. When there is a short at the output during operation, the current limit is decreased accordingly.

An under voltage lockout function prevents the main output at VOUT from being supplied if the voltage at UVLO drops below 0.25 V. When using a resistive divider at the voltage to be monitored, for example the supply voltage, any threshold for the monitored voltage can be programmed. If in under voltage lockout mode, the device still maintains its supply voltage at VAUX, but it is not turned off until EN is programmed low. This under voltage lockout function is implemented in order to prevent the malfunctioning of the converter. The device has a built-in temperature sensor which monitors the internal IC temperature. If the temperature exceeds the programmed threshold (see electrical characteristics table), the device stops operating. As soon as the IC temperature has decreased below the programmed threshold, it starts operating again. There is a built-in hysteresis to avoid unstable operation at IC temperatures at the over temperature threshold.

To properly configure the fixed output voltage devices, the FB pin is used to sense the output voltage. This means that it must be connected directly to VOUT. At the adjustable output voltage versions, an external resistor divider is used to adjust the output voltage. The resistor divider must be connected between VOUT, FB and GND. When the output voltage is regulated properly, the typical value of the voltage at the

FB pin is 500 mV. The maximum recommended value for the output voltage is 5.5 V. The current through the resistive divider should be about 100 times greater than the current into the FB pin. The typical current into the FB pin is 0.01 uA, and the voltage across the resistor between FB and GND, R2, is typically 500 mV. Based on those two values, the recommended value for R2 should be lower than 500 kΩ, in order to set the divider current at 1uA or higher. It is recommended to keep the value for this resistor in the range of 200 kΩ. The value of the resistor connected between VOUT and FB, R1, depending on the needed output voltage (VOUT), can be calculated using following equation:

$$R1 = R2 \times \left\{ \frac{V_{out}}{V_{FB}} - 1 \right\} \quad (\text{Equation 2})$$

Since the typical voltage range of the digital components is less than 3.6 volts to achieve the 3.3 Volts in output, R1 = 1.2MΩ and R2 = 220KΩ has been calculated. The UVLO input can be used to shut down the main output if the supply voltage is getting too low. The internal reference threshold is typically 250 mV. If the supply voltage should cause the shutdown when it is dropping below 250 mV, it can be connected directly to the UVLO pin. If the shutdown has already happen at higher voltages, a resistor divider can be used. The purpose of R3 and R4 shown in Fig 18 as a voltage divider are to monitor the voltage of the circuit. The current through the resistive divider should be about 100 times greater than the current into the UVLO pin. The typical current into the UVLO pin is 0.01 A, and the voltage across R4 is equal to the UVLO voltage threshold that is generated on-chip, which has a value of 250 mV. The recommended value for R4 is; therefore, in the range of 250 kΩ. From this, the value of resistor R3, depending on the desired shutdown voltage VIN MIN, can be calculated using the following equation.

$$R8 = R9 \times \left\{ \frac{V_{inmin}}{V_{ulvlo}} - 1 \right\} \quad (\text{Equation 3})$$

To avoid reducing Li-Po Battery life by dropping the battery voltage under 2.5 volts threshold voltage has been set to 2.5 volts by calculating the $R8 = 2M\Omega$ and $R9 = 220K\Omega$. To make sure that the TPS6120X devices can operate, an inductor must be connected between pin VIN and pin L. To estimate the minimum inductance value:

$$L = V_{min} \times 0.5 \frac{\mu s}{A} \quad (\text{Equation 4})$$

The recommended inductor value range is between 1.5uH and 4.7uH. The minimum inductor value should not be below 1.5uH, even if Equation 4 yields in something lower. To achieve the best performance 4.7uH is has been chosen for L1 inductor. To improve transient behavior of the regulator and EMI behavior of the total power supply circuit value of 100uf has been selected for input capacitor C8. Also for output capacitor small ceramic capacitors placed as close as possible to the VOUT and PGND pins of the IC.

Between the VAUX pin and GND a 0.1uf capacitor has been connected. This capacitor is used to maintain and filter the control supply voltage. It is charged during startup and before the main output VOUT is turned on. To ensure stable operation, using at least 0.1F is recommended. At output voltages below 2.5 V, the capacitance should be in the range of 1 F. Since this capacitor is also used as a snubber capacitor for the main switch, using a ceramic capacitor with low ESR is important.

Implementation of integrated circuits in low-profile and fine-pitch surface-mount packages typically requires special attention to power dissipation. Many system-dependent issues such as thermal coupling, airflow, added heat sinks and

convection surfaces, and the presence of other heat-generating components affect the power-dissipation limits of a given component.

Three basic approaches for enhancing thermal performance are listed below:

Improving the power dissipation capability of the PCB design.

Improving the thermal coupling of the component to the PCB.

Introducing airflow in the system.

The maximum recommended junction temperature (T_J) of the TPS6120x devices is 125°C. Specified regulator operation is assured to a maximum ambient temperature T_A of 85°C. Therefore, the maximum power dissipation is about 820 mW [27]. More power can be dissipated if the maximum ambient temperature of the application is lower. As it shown in Fig 19, the underneath ground plate of the IC is connected to the large ground polygon at the bottom layer of the board with nine via holes. This approach causes all heats radiated from the IC dissipates easily to ground polygons and keeps the junction temperature very much less than limits.

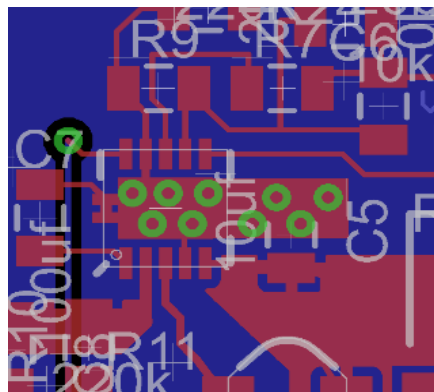


FIGURE 19. Boost converter heat dissipation approach.

Microcontroller and peripherals. With selecting the microcontroller having low power consumption, it is also necessary to choose appropriate peripherals (GPIO, ADC, I2C, USB, Serial, etc...), digital signal processing capabilities, open source development environment and strong support from developer community.

To meet all these requirements, we have selected LPC1549 microcontroller. The LPC15xx are ARM Cortex-M3 based microcontrollers for embedded applications featuring a rich peripheral set with very low power consumption. The ARM Cortex-M3 is a next generation core that offers system enhancements such as enhanced debug features and a higher level of support block integration. The LPC15xx operates at CPU frequencies of up to 72 MHz. The ARM Cortex-M3 CPU incorporates a 3-stage pipeline and uses Harvard architecture with separate local instruction and data buses as well as a third bus for peripherals. The ARM Cortex-M3 CPU also includes an internal pre-fetch unit that supports speculative branching. The LPC15xx include up to 256 kB of flash memory, 32 kB of ROM, a 4 kB EEPROM, and up to 36 kB of SRAM. The peripheral complement includes one full-speed USB 2.0 device, two SPI interfaces, three USARTs, one Fast-mode Plus I2C-bus interface, one C_CAN module, PWM/timer subsystem with four configurable, multi-purpose State Configurable Timers (SCTimer/PWM) with input pre-processing unit, a real-time clock module with independent power supply and a dedicated oscillator, two 12-channel/12-bit, 2M samples/sec ADCs, one 12-bit, 500k Samples/sec DAC, four voltage comparators with internal voltage reference, and a temperature sensor. A DMA engine can service most peripherals [28].

The ARM Cortex-M3 includes three AHB-Lite buses, one system bus and the I-code and D-code buses. One bus is dedicated for instruction fetch (I-code), and one

bus is dedicated for data access (D-code). The use of two core buses allows for simultaneous operations if concurrent operations target different devices. A multi-layer AHB matrix connects the Cortex-M3 buses and other bus masters to peripherals in a flexible manner that optimizes performance by allowing peripherals on different slave's ports of the matrix to be accessed simultaneously by different bus masters. APB peripherals are connected to the CPU via two APB buses using separate slave ports from the multilayer AHB matrix. This allows for better performance by reducing collisions between the CPU and the DMA controller. The APB bus bridges are configured to buffer writes so that the CPU or DMA controller can write to APB devices without always waiting for APB write completion.

The ARM Cortex-M3 is a general purpose 32-bit microprocessor, which offers high performance and very low power consumption. The Cortex-M3 includes a Thumb-2 instruction set and provides low interrupt latency, hardware divide, interruptible/continuable multiple load and store instructions, automatic state save and restore for interrupts, tightly integrated interrupt controller, and multiple core buses capable of simultaneous accesses.

Pipeline techniques are employed so that all parts of the processing and memory systems can operate continuously. Typically, while one instruction is being executed, its successor is being decoded, and a third instruction is being fetched from memory [28]. For designing of the board, to be able to use all feature of the MBED open source software stack API, all pin-outs mapping adopted from a MBED compatible development board [29].

GPIO's. The GPIO pins can be used in several ways to set pins as inputs or outputs and use the inputs as combinations of level and edge sensitive interrupts. GPIOs can be used as external interrupts together with the pin interrupt and group

interrupt blocks. The GPIO port registers configure each GPIO pin as input or output and read the state of each pin if the pin is configured as input or set the state of each pin if the pin is configured as output.

PIO_25-BREAK_CTRL-RED. This GPIO in Fig 20 controls an SMD led which can be programmatically controlled as status indicator as can be seen in the Fig 20 anode side of the LED connected to +3.3 with a 1.5k Ω to limit the current flow.

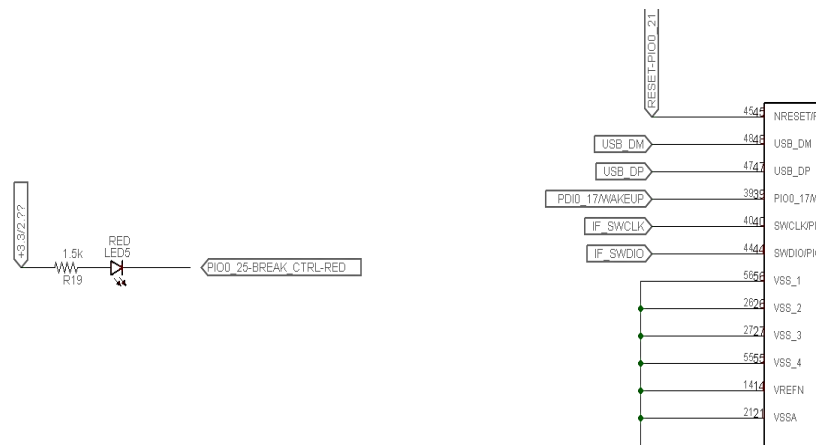


FIGURE 20. PIO_25-BREAK_CTRL-RED LED control GPIO.

PIO1_3/ADC1_5. To turn on/off the Bluetooth module or reset it in case of any malfunction, I/O is connected to RESET pin of the RN42 (shown in Fig 21).

The RN42 Reset pin is active low. As shown in Fig 22, this pin is pulled up with 1.5K Ω resistor. It means the radio will be turned off until the firmware turns the PIO1_3 to low. It approaches keeps the radio off during the MCU's boot time to avoid any fast voltage drop down or any high frequency noise interference.

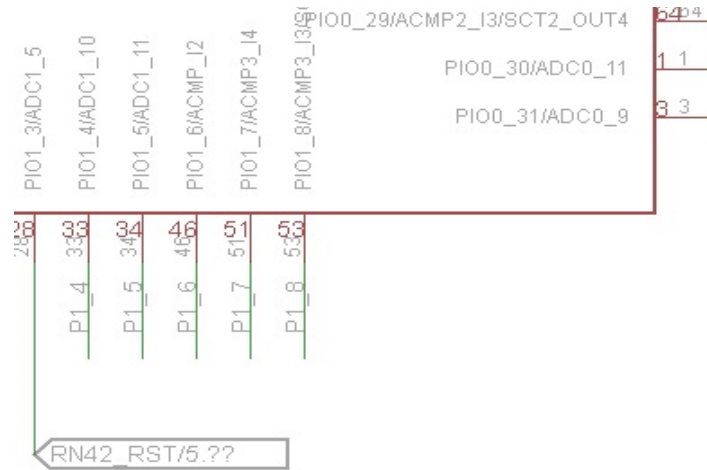


FIGURE 21. RN42 reset control GPIO.

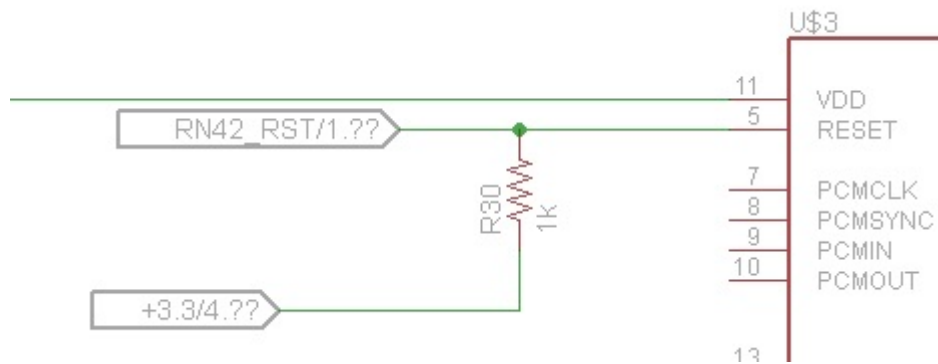


FIGURE 22. RN42 reset Pin PullUp.

Micro switches. There is 4 switches S1,S2,S3,S4 assigned to GPIO's for different purposes. S1 is assigned to PIO1_9, and S2 is connected to PIO_11 (shown

in Fig 23). Both can be programmatically connected to fire a GPIO external interrupt as user controls which each of them internally used to run user defined specific tasks.

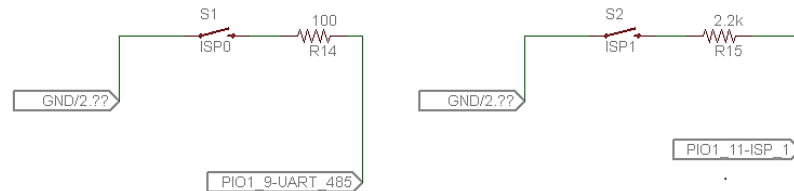


FIGURE 23. Micro switches S1,S2.

S2 shown in Fig 23 is connected to reset pin of MCU and it is not usable as user input. This switch is assigned specifically to rest or boot select of the MCU. As it is illustrated in Fig 24, reset pin PIO0_21 is pulled up to +3.3 volts with R2 to keep reset pin always high. Capacitor C20 is an EMI bypass cap to avoid any unwanted reset with noise spike on the line. By pressing the S4, PIO0_21 will pull down through R4 to the ground and will cause MCU reset while it will complete the current path for LED2 and it will turn on when the reset pin has been pressed.

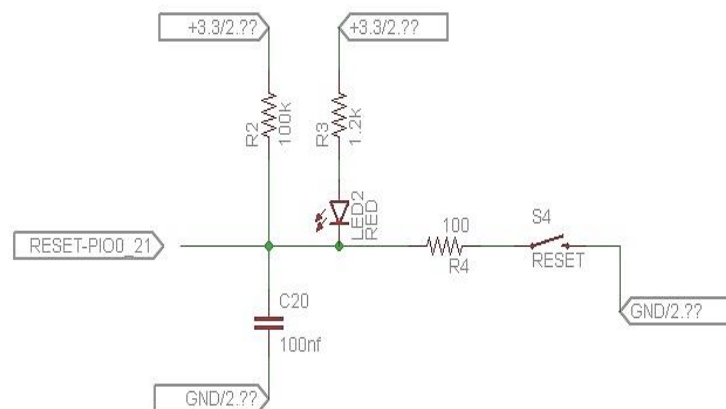


FIGURE 24. MCU's reset switch.

By combining RESET button, S1, and S2 it is possible to select boot loader channel after reset.

Analog to digital. For sampling the analog ECG signal and pre-signal conditioning LPC 1549's internal analog to digital converter has been used. Also another input assigned to sample the battery voltage which allows developers to monitor battery usage and power efficiency. LPC1549's ADC features with 12-bit successive approximation analog-to-digital converter. It is capable of 12-bit conversion rate of 2 MHz Input multiplexing among 12 pins and up to 4 internal sources. It can measure from VREFN to VREFP (typically 3 V; not to exceed VDDA voltage level). As illustrated in Fig 25, while VREFN is connected to ground, VREFP is connected to 3.0V Which is provided with highly stable reference voltage chip, LT1790 [30]. C25, C24, and C23 in together work as a bypass capacitors network to eliminate EMI high and midrange frequency noises by bypassing to the ground. These capacitors are placed as closest as possible to the MCU. It means there is shortest surface of a trace from capacitors to MCU, therefore there is less surface exposed to potential EMI noises.



FIGURE 25. VREF bypass capacitors.

To be able to measure VBAT, voltage of Li-Po battery with ADC a voltage divider with two resistors R24 and R25 divides in Fig 26. The battery voltage with 1/10 rate. Since our reference voltage is 3.00 volts ADC unit cannot measure voltages more than 3 volts. While the battery voltage swings between minimum 2.5Volts to fully charged 4.2volts, to be able to measure the VBAT 1/10 dividing range guarantees that ADC will cover full range of measurement. To increase the accuracy of voltage divider resistors with 1% tolerance has been placed. The output of voltage divider is connected to P0_8 which is also adjustable as ADC0's first input (ADC0_0).

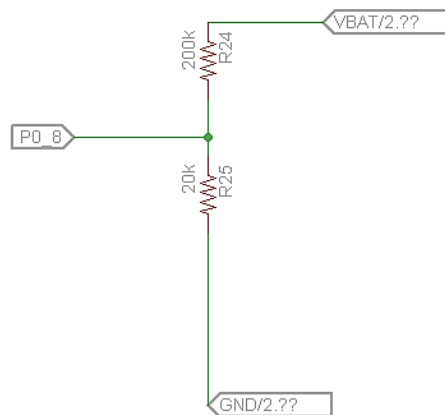


FIGURE 26. VBAT voltage divider by 1/10.

For measuring the ECG board analog output, P0_7/ADC0_1 Second input of analog input 0 is linked to ECG board by board to board 5*2 FTSH-1050DV box type pin-headers Fig. 27.

may choose to optimize range to 200 feet and beyond, particularly for in-home sensor applications where longer range is a necessity.

Bluetooth Smart features provides:

- Ultra-low peak, average and idle mode power consumption

- Ability to run for years on standard coin-cell batteries

- Lower implementation costs

- Multi-vendor interoperability

- Enhanced range

This enhancement to the Bluetooth Core Specification allows two types of implementation, dual-mode and single-mode. In a dual-mode implementation, Bluetooth low energy functionality is integrated into an existing Classic Bluetooth controller. The resulting architecture shares much of Classic Bluetooth technology's existing radio and functionality resulting in a minimal cost increase compared to Classic Bluetooth technology. Additionally, manufacturers can use current Classic Bluetooth technology (Bluetooth v2.1 + EDR or Bluetooth v3.0 + HS) chips with the new low energy stack, enhancing the development of Classic Bluetooth enabled devices with new capabilities. Single-mode chips, which will enable highly integrated and compact devices, will feature a lightweight Link Layer providing ultra-low power idle mode operation, simple device discovery, and reliable point-to-multipoint data transfer with advanced power-save and secure encrypted connections at the lowest possible cost. The Link Layer in these controllers will enable Internet connected sensors to schedule Bluetooth low energy traffic between Bluetooth transmissions. Experimental results in Fig 28. Shows BLE consumes even less than ZigBee and ANT for same amount of data traffic amount [31].

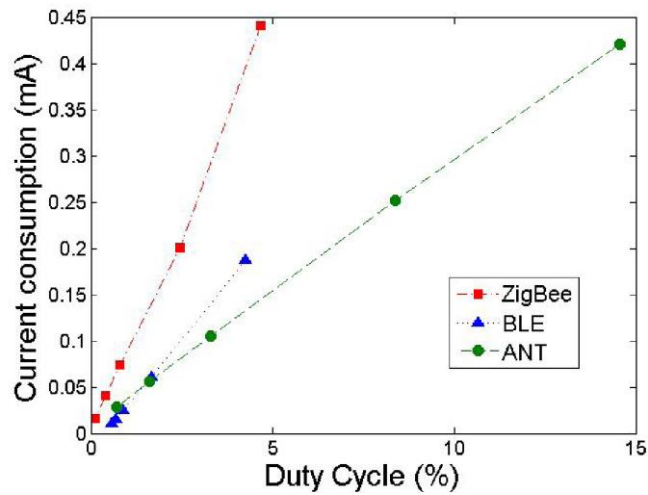


FIGURE 28. Power consumption comparison between ZigBee, BLE, ANT.

As it is shown in Table 4, power consumption efficiency of BLE in experimental test with 3.3 volts supply is not comparable.

TABLE 4. BLE, ZigBee, ANT Power Consumption Values

	BLE	ZigBee	ANT
Time of one Connection	115ms ±260ms	250ms ±9.1ms	930ms ±230ms
Sleep current	0.78uA	4.18uA	3.1uA
Awake Current	4.5mA	9.3mA	2.9mA
Min current (at 120 sec interval)	10.1uA	15.7uA	28.2uA
Optimal sleep interval	10.0s	14.3s	15.3s

These characteristics of BLE made it as the best candidates for wearable medical platform. For these reasons initially tests has been done with CC2541 from Texas Instruments. This chip comes with complete Power-Optimized Stack, Including

Controller and Host. The stack supports GAP – Central, Peripheral and Observer, ATT / GATT – Client and Server and also SMP AES-128 Encryption. But the firmware development environment is not free or open source and it costs couple of thousand dollars to buy it which not reasonable for researchers. Besides, communicating with BLE is not straightforward as standard Bluetooth. Most of the PC's are not supporting the BLE. Even if the hardware supports BLE desktop operating system does not have infrastructure to communicate Low Power Bluetooth profiles. Since researchers need standard, easy to develop and commonly available data link for further data analysis in later version of the design, instead of BLE, standard version 2.1 Bluetooth, The RN-42 module has been replaced with BLE. This module supports standard Serial Port Profile "SPP" and also Human Interface Device "HID" which means using third party software's for data analysis and digital signal processing such as MATLAB will be much easier.

The RN-42 is a small form factor, low power, class 2 Bluetooth radio for designer's who want to add wireless capability to their products. The RN-42 supports multiple interface protocols, is simple to design in, and is fully certified, making it a complete embedded Bluetooth solution. With its high-performance, on-chip Antenna and support for Bluetooth EDR, the RN-42 delivers up to a 3 Mbps data rate for distances up to 20 meters.

To compare power consumption between these two different approaches, clearly RN-42 consumes more than CC2541.

RN-42 consumes 45mA while radio is in transmitting mode [32]. while CC2541 consumes 20mA. This trade-off makes sense by considering ease of use and adding more flexibility to platform for further development.

Fig 29. is the schematic of the implemented Bluetooth section.

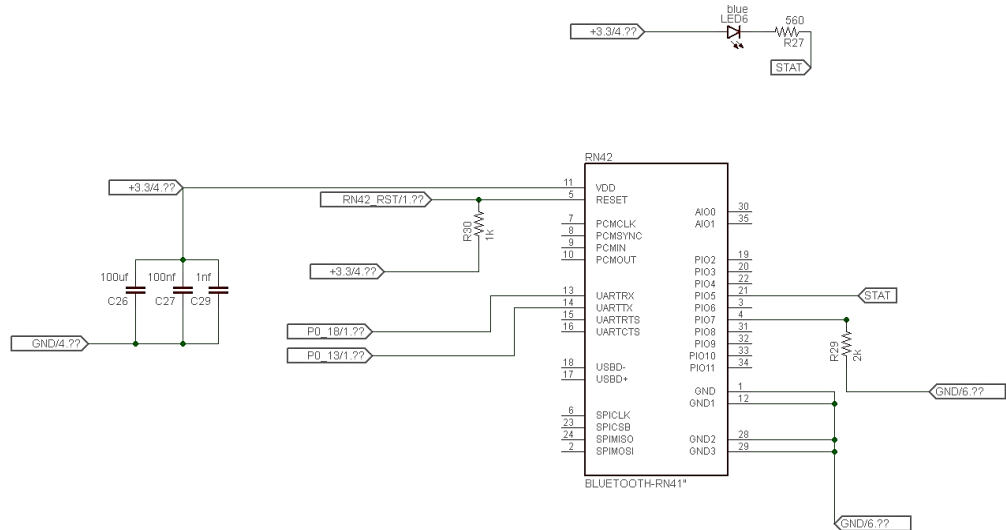


FIGURE 29. RN-42 schematic.

UART connections of the RN-42 is connected to MCU's UART number zero. All acquired data from sensors and analog inputs stream to through this communication channel to RN-42. RN-42's firmware defined in SPP mode by default which means by pairing this board with any computer, smart phone or tablet it will add a virtual serial port over Bluetooth to the list of the port. It means the host application can open that specific port and start sends commands or receives data streams.

Debug

Debug functions are integrated into the ARM Cortex-M3. Serial wire debug functions are supported in addition to a standard JTAG boundary scan. The ARM Cortex-M3 is configured to support up to four breakpoints and two watch points.

The RESET pin selects between the JTAG boundary scan (RESET = LOW) and the ARM SWD debug (RESET = HIGH). The ARM SWD debug port is disabled while the LPC15xx is in reset.

To perform boundary scan testing, follow these steps:

1. Erase any user code residing in flash.
2. Power up the part with the RESET pin pulled HIGH externally.
3. Wait for at least 250 s.
4. Pull the RESET pin LOW externally.
5. Perform boundary scan operations.
6. Once the boundary scan operations are completed, assert the TRST pin to enable the SWD debug mode, and release the RESET pin (pull HIGH).

Normally all these steps are handled by the IDE and the Debugger such as LPC Link, JTAG from SEGGER, ULINK from KEIL.

Since, JTAG was the traditional mechanism for debug connections for ARM7/9 parts, but with the Cortex-M family, ARM introduced the Serial Wire Debug (SWD) Interface. SWD is designed to reduce the pin count required for debug from the 5 used by JTAG (including GND) down to 3. In addition, one of the pins freed up by this can be used for Single Wire Viewing (SWV), which is a low cost tracing technology (which is used by the "Red Trace" functionality within Red Suite). Usually, MCUs do not include pull-up or pull-down resistors on JTAG/SWD pins. Resistors should be added externally onto the board as detailed above. In this design 10k resistors placed for these signals. This will prevent the signals from floating when they are not connected to anything.

As it is illustrated in Figure 30. standard 10 pin FTSH-105-DV connector selected to match standard debuggers.

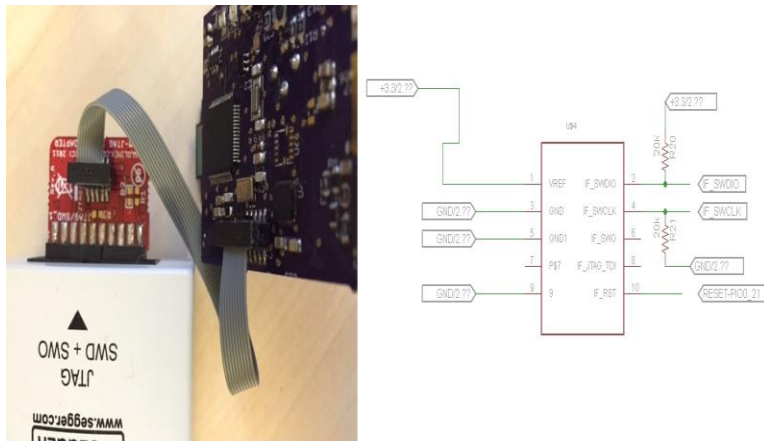


FIGURE 30. JTAG debugger Connection.

Implemented Sensors

Temperature. Body temperature abnormalities, which occur because of several infectious and noninfectious etiologies, are among the most commonly noted symptoms of ill patient. A reliable wearable medical gadget with medical accuracy continues temperature monitoring can be very useful because lot of serious body abnormality might cause fever (high temperature) or hypothermia (low temperature). Normally classic medical temperature measurement techniques such as Liquid-in-glass, thermocouple, thermistors and Integrated Circuit temperature sensors (IC thermometer) need to be in touch with body for a while to absorb the body temperature and be at the same temperature heat which means user should keep the device in touch with the body for a while to get the accurate results. But Since in the wearable device which the position of the device might not be very tight to the body and Also users move during the daily activities all of mentioned methodologies cannot be an option except Radiation thermometers and other non-contact thermometers. Each body with a temperature above the absolute zero ($-273.15^{\circ}\text{C} = 0$ Kelvin) emits an electromagnetic radiation from its surface, which is proportional to

its intrinsic temperature. A part of this so-called intrinsic radiation is infrared radiation, which can be used to measure a body's temperature [33].

This radiation penetrates the atmosphere. With the help of a lens (input optics) the beams are focused on a detector element, which generates an electrical signal proportional to the radiation. The signal is amplified and, using successive digital signal processing, is transformed into an output signal proportional to the object temperature. Which means Infrared temperature sensors can sample very fast even from a proper distance with a proper optic front end, medical rated type of MLX90614 Infrared sensor selected.

The MLX90614 is an Infrared thermometer for non-contact temperature measurements. Both the IR sensitive thermopile detector chip and the signal conditioning ASS Pare integrated in the same TO-39 can. Thanks to the low noise amplifier, high resolution 17-bit ADC and powerful DSP unit of MLX90614 high accuracy and resolution of the thermometer is achieved. The calculated object and ambient temperatures are available in RAM of MLX90614 with resolution of 0.01°C. They are accessible by 2 wire serial SMBus compatible protocol (0.02°C resolution) or via 10-bit PWM (Pulse Width Modulated) output of the device. The MLX90614 is factory calibrated in wide temperature ranges: -40...125°C for the ambient temperature and -70...380°C for the object temperature. The measured value is the average temperature of all objects in the Field Of View of the sensor. The MLX90614 offers a standard accuracy of $\pm 0.5^{\circ}\text{C}$ around room temperatures. A special version for medical applications (which has been used in this design) has an accuracy of $\pm 0.2^{\circ}\text{C}$ in a limited temperature range around the human body temperature as illustrated in Fig. 31[34].

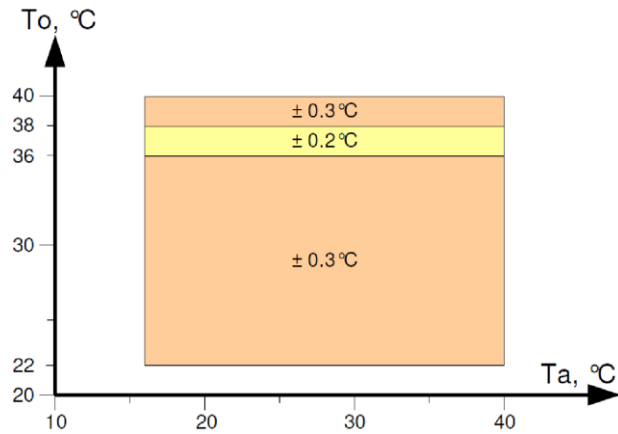


FIGURE 31. MLX90614 temperature accuracy in body temperature range.

As we can see sensor has a very precise accuracy in human body temperature range which makes it valuable fast measurement tool. To make sure body is close enough to be in field of view of the sensor placed in upper digital board, bottom layer looking to the skin of user. As it is illustrated in Fig 32. a proper hole in front of the sensor in lower analog board has been placed also.

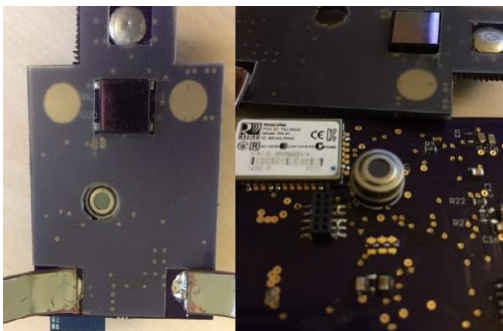


FIGURE 32. Temperature sensor placement.

Motion processor unit. Nowadays, commercial activity, movement trackers paired smart phones are very popular. These devices have the capability of monitoring amount of daily physical activity, sleep quality, fall detection, walking pattern abnormality and stress to the user. Having a history of these data beside other vital sign in long term might be useful for further abnormality analysis and causes related to them which itself can be a sophisticated long term research project to help users to improve their health by feedbacks provided by the personalized back end big data analysis. For this purpose MPU-9150 from InvenSense has been selected.

The MPU-9150 is the world's first integrated 9-axis MotionTracking device that combines a 3-axis MEMS gyroscope, a 3-axis MEMS accelerometer, a 3-axis MEMS magnetometer and a Digital Motion Processor™ (DMP™) hardware accelerator engine. The MPU-9150 is an ideal solution for handset and tablet applications, game controllers, motion pointer remote controls, and other consumer devices. The MPU-9150's 9-axis MotionFusion combines acceleration and rotational motion plus heading information into a single data stream for the application. This MotionProcessing™ technology integration provides a smaller footprint and has inherent cost advantages compared to discrete gyroscope, accelerometer, plus magnetometer solutions. The MPU-9150 is also designed to interface with multiple non-inertial digital sensors, such as pressure sensors, on its auxiliary I2C port to produce a 10-Axis sensor fusion output. The MPU-9150 is a 3rd generation motion processor and is footprint compatible with the MPU-60X0 and MPU-30X0 families. The MPU-9150 features three 16-bit analog-to-digital converters (ADCs) for digitizing the gyroscope outputs three 16-bit ADCs for digitizing the accelerometer outputs and three 13-bit ADCs for digitizing the magnetometer outputs. For precision tracking of both fast and slow motions, the parts feature a user programmable

gyroscope full-scale range of ± 250 , ± 500 , ± 1000 , and $\pm 2000^\circ/\text{sec}$ (dps), a user programmable accelerometer full-scale range of $\pm 2g$, $\pm 4g$, $\pm 8g$, and $\pm 16g$, and a magnetometer full-scale range of $\pm 1200\mu\text{T}$ [35].

The MPU-9150 is a multi-chip module (MCM) consisting of two dies integrated into a single LGA package. One die houses the 3-Axis gyroscope and the 3-Axis accelerometer. The other die houses the AK8975 3-Axis magnetometer from Asahi Kasei Microdevices Corporation. An on-chip 1024 Byte FIFO buffer helps lower system power consumption by allowing the system processor to read the sensor data in bursts and then enter a low-power mode as the MPU collects more data. With all the necessary on-chip processing and sensor components required to support many motion-based use cases, the MPU-9150 uniquely supports a variety of advanced motion-based applications entirely on-chip.

The MPU-9150 thus enables low-power MotionProcessing in portable applications with reduced processing requirements for the system processor. By providing an integrated MotionFusion output, the DMP in the MPU-9150 offloads the intensive MotionProcessing computation requirements from the system processor, minimizing the need for frequent polling of the motion sensor output.

Communication with all registers of the device is performed using I2C at 400kHz. Additional features include an embedded temperature sensor and an on-chip oscillator with $\pm 1\%$ variation over the operating temperature range.

By leveraging its patented and volume-proven Nasiri-Fabrication platform, which integrates MEMS wafers with companion CMOS electronics through wafer-level bonding, InvenSense has driven the MPU-9150 package size down to a revolutionary footprint of $4 \times 4 \times 1\text{mm}$ (LGA), while providing the highest performance, lowest noise, and the lowest cost semiconductor packaging required for handheld

consumer electronic devices. The part features a robust 10,000g shock tolerance, and has programmable low-pass filters for the gyroscopes, accelerometers, magnetometers, and the on-chip temperature sensor. [35].

The embedded Digital Motion Processor (DMP) is located within the MPU-9150 and offloads computation of motion processing algorithms from the host processor. The DMP acquires data from accelerometers, gyroscopes, magnetometers and additional 3rd party sensors such as pressure sensors, and processes the data. The resulting data can be read from the DMP's registers, or can be buffered in a FIFO. The DMP has access to one of the MPU's external pins, which can be used for generating interrupts.

The purpose of the DMP is to offload both timing requirements and processing power from the host processor. Typically, motion processing algorithms should be run at a high rate, often around 200Hz, in order to provide accurate results with low latency. This is required even if the application updates at a much lower rate; for example, a low power user interface may update as slowly as 5Hz, but the motion processing should still run at 200Hz. The DMP can be used as a tool in order to minimize power, simplify timing, simplify the software architecture, and save valuable MIPS on the host processor for use in the application.

Also, with this sensor developers will be able to detect in what situation the other sensors has been recorded. For example it will distinguishable if the temperature data has been recorded while user was sleeping, sitting or running. It means it will be possible to analyze data of user based in each activity to reduce analysis failures.

Proximity sensor. Vcnl4000, Proximity sensor is useful to detect if the device is in use or not, besides this specific sensor can measure environment ambient light. The ambient light record can help developers to distinguish day or night activates.

The VCNL4000 is a proximity sensor with an integrated ambient light sensor. It is the industry's first optical sensor to combine an infrared emitter, PIN photodiode, ambient light sensor and signal processing in a single package with a 16-bit ADC for proximity measurement as well as ambient light measurement. The device provides ambient light sensing to support conventional backlight and display brightness adjustment, and proximity sensing for object and motion detection. With a range of up to 20 cm (7.9"), this stand-alone, single component greatly simplifies the use and design-in of a proximity sensor in consumer and industrial applications. The VCNL4000 features a miniature leadless package (LLP) for surface mounting in a 3.95 mm x 3.95 mm package with a low profile of 0.75 mm is a good benefit for compact designs like wearable devices. Through its standard I2C bus serial digital interface, it allows easy access to a "Proximity Signal" and "Light intensity" measurement without complex calculations or programming [36].

Analog Hardware Design Details and Implementation

Electrocardiograph

Electro Cardio Gram (ECG) is a test that is used to check for electrical activities of the heart [37]. The heart activities are translated into either a tracing paper or on a real time monitoring screen. These traces are the small voltage signal generated from the heart due to contraction and relaxation of the heart. The spikes and dips in the traces are referred to as waves [38]. Placing the leads on specific position on the body different heart conditions can be learned by looking at the characteristics of the ECG wave.

An ECG signal is generated from a single heartbeat. Every one heartbeat generates one ECG signal. A normal heart beat would generate all the ECG waves namely, P wave, QRS complex, and the T wave. In some cases, an ECG signal could

have a U wave. The Fig 33. shows a normal ECG generated from a single heart beat where all the waves are clearly shown.

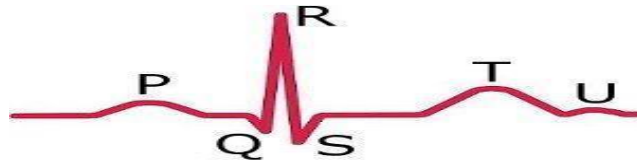


FIGURE 33. Proper ECG Signal.

As stated earlier, an ECG signal is generated from the electrical activity of the heart. The heart consists of four chambers. The top chamber is called the atrium and consists of the left and right atrium. The bottom chamber is called the ventricle and is divided into the left and right ventricle. The heart is made up of muscles called myocardium and the contraction/relaxation of these muscles lead to the flow of blood and thus the generation of the ECG wave [39].

The electrical impulse generation starts from the right atrium of the heart in the sinoatrial node (SA node). This tissue is responsible for the generation of the electrical impulse. The impulse generated travels down and leads to the contraction of the right atrium. At the same time, the impulse travels to the left atrium and causes the contraction of the left atrium as well. The contraction of the top chamber of the heart causes the blood to be pumped out and thus the formation of the P wave [39].

The impulse then travels down to the atrioventricular node (AV node). This tissue connects both the top and bottom chambers of the heart together. As the

impulse gets to this node there is a little bit of delay and thus generates the flat segment between the P and Q wave.

From the AV node the impulse travels down to the bundle of His, which has branches that spread out to the bottom chamber of the heart. These branches are connected with Purkinje fibers which extend to the left and right walls of the ventricle. As the impulse reaches these muscles from the bundle of His branches and through the Purkinje fibers, both the left and right ventricle contracts and this contraction leads to blood be pumped out of the heart and thus causes the QRS complex.

As it can be seen the P wave is smaller than the QRS complex because the muscles of the atrium are smaller than the muscles of the ventricles and thus the electrical charges that lead to contraction of the atrium are smaller compared to the ones that lead to the contraction of the ventricles.

As the ventricle relax after ejecting blood from the heart, the electrical impulses that lead to the relaxation of the ventricles cause the formation of the T wave [39]. In some cases the U wave can be seen in an ECG signal and this is caused by the relaxation of the bundle of His and Purkinje fibers.

An ECG Signal has a wide variety of application in industries especially in medical field. Some of the many applications of the ECG signals include the measurement of the heartbeat, and diagnostic purposes [40].

In the measurement of the heartbeat, there are different ways the signal could be utilized to achieve that purpose. One of the ways to measure the heartbeat of a person is to record the ECG signal in a period of six seconds. In this interval, one counts the number of QRS peaks available and multiplies that number by ten. The result gives an approximate heartbeat of a person.

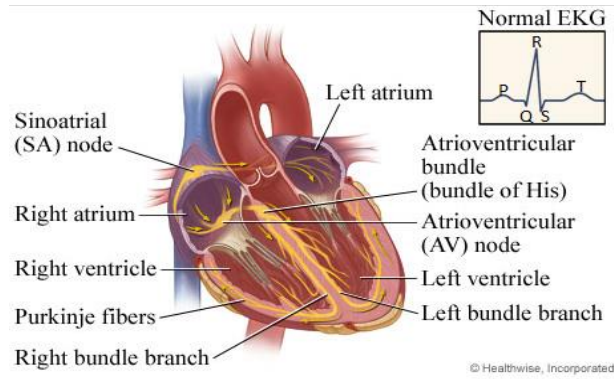


FIGURE 34. Impulse path in a healthy heart.

For instance in the Fig 35. the total number of QRS peak in a six second interval is seven peaks. This number is multiplied by ten to give 70 beats per minutes.



FIGURE 35. ECG of a healthy person.

Another application of ECG is in diagnostics. The ECG signals obtained from a person can be used to check for different kinds of heart condition. Each of the wave in an ECG signal represent functionality of different part of the heart as described above. If any of the wave obtained is deviated from the normal ECG wave then that information could guide the physician to diagnose the patient with a condition of the heart that could lead to that deviation.

Some examples of a heart condition that could be noted from ECG signals are sinus Bradycardia and sinus tachycardia [40]. The heartbeat of a normal person is between 60 – 100 beats per minute. Sinus Bradycardia is a heart condition that occurs when the number of beat is less sixty in a minute. And Sinus tachycardia is heart condition that occurs when the heartbeat of a person is more 100 beats per minute.

Similarly to the heartbeat measurement, when an ECG signal is observed in a six seconds interval and the QRS complex in that interval is less than six then the person could be diagnosed with sinus Bradycardia. Similarly, when the number of QRS complex in a six seconds interval is more than 10 then the person could be diagnosed with Sinus tachycardia and looked for further treatment.

Hardware. Generating the ECG signal significantly depends on the accuracy of hardware design. There are many aspects that should be concerned in designing the electronics to collect the appropriate signals from the body and convert them to the accurate ECG signal. An overview of the hardware section is shown in Fig 36.

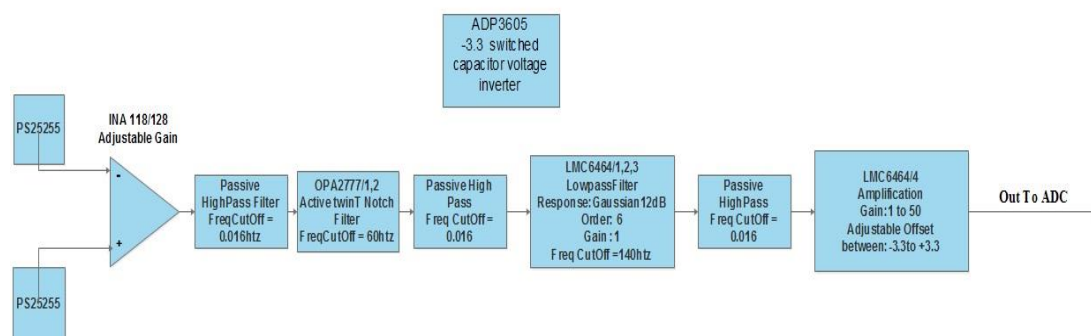


FIGURE 36. Hardware overview.

Clearly, the most important part of hardware design are the Electrodes. Electrodes are the main components to get feedback from the body condition. Based

on the explanations in the previous paragraphs, we are looking for the changes of body skin charges which correspond to the ECG signal. As a result we are looking for Electrodes which can monitor such condition of the body. There are many types of ECG Electrodes for collecting appropriate signals from body including dry contact ECG sensor, non-dry contact ECG electrodes, and non-contact ECG electrodes. Conventionally in many medical monitoring conditions in hospitals, the non-dry contact ECG electrodes are used to generate the ECG signal. The drawback of using such electrodes is that we have to use conductive gel with these sensors to assure the connectivity of the electrode and the body skin. On the other side, there are dry contact ECG active electrodes which are used in this project. The PS25255 is an ultra-high impedance solid state ECG (electrocardiogram) sensor. It can be used as a dry contact ECG sensor without the need for potentially dangerous low impedance circuits across the heart.

The resolution available is as good as or better than conventional wet electrodes. The device uses active feedback techniques to both lower the effective input capacitance of the sensing element (C_{in}) and boost the input resistance (R_{in}). These techniques are used to realize a sensor with a frequency response suitable for both diagnostic and monitoring ECG applications. The internal design of this sensor is shown in Fig 37. [41]

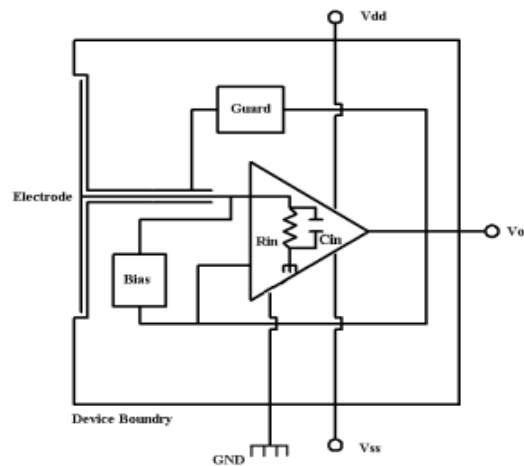


FIGURE 37. PS25255 ECG sensor schematic.

In Portable/wearable ECG platform it is very essential to avoid any types of non-dry electrodes. Since the goal is sustain data acquisition continuously in most convenient way, the cutting edge approach of PS25255 from Plessey Semiconductors has been chosen for this purpose.

The supply Voltage of the sensors are between $\pm 2.4\text{v}$ to $\pm 5.5\text{v}$ with ground on Zero volt. Since in this design we just have single $+3.3\text{ volts}$ supply, the ADP3605 switched capacitor voltage inverter has been used to provide -3.3 volts for analog front end. It provides a regulated output voltage with minimum voltage loss and requires a minimum number of external components. In addition, the ADP3605 does not require the use of an inductor.

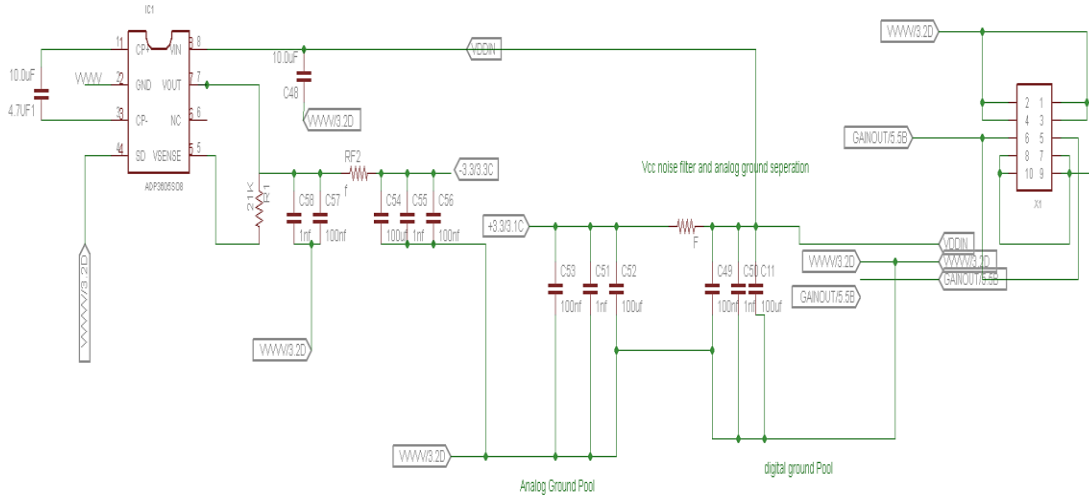


FIGURE 38. ADP 3605 voltage inverter schematic.

The ADP3605 provides an accuracy of $\pm 3\%$ with a typical shutdown current of $2 \mu\text{A}$. It can also operate from a single positive input voltage as low as 3 V .

The ADP3605 is adjustable via external resistors over a -3 V to -6 V range. The schematic is illustrated in Fig 38 [42].

The ADP3605 to reach to high accuracy on the adjustable output voltage is achieved with the use of precision trimmed internal resistors, which eliminate the need to trim the external resistor or add a second resistor to form a divider. The adjustable output voltage is set by

$$V_{OUT} = \frac{1.5}{K\Omega} R$$

Where V_{OUT} is in volts and R is in $\text{k}\Omega$. As a result, -3.315 volts has been achieved by calculating $21\text{k}\Omega$ for $R1$.

Noise can be easily introduced by the digital high frequency components to analog and affect the quality of the analog signal through the same ground plane. To avoid this phenomenon, since we have one ground for both digital and analog sections and reduce EMI noises also the analog ground has been separated by 3 order of

bypass capacitors. The lower value capacitor shunts high-frequency noise coupled on the power supply traces to ground due to its low impedance in the higher frequency range [43]. Also the inductor (ferrite beads) “F” will absorb noise spikes of VCC before distributed in analog board.

Instrumentation amplifier with INA128. The ECG potential is not the only signal found at the input of the amplifier; one major source of interference is the electrical power system. Capacitance between power lines in the wall, floor and ceiling and nearby equipment couples current into the patient, wires and machine. This current flows through the skin-electrode impedances on the way to ground. The capacitance to these power line sources varies with proximity but is of the order of 50pF which corresponds to an impedance of 64M Ω at 60Hz [44]. The right leg electrode (which in this design is the user wrist instead) is connected to the common ground of the amplifier through an electrode with contact impedance of, say, 5k Ω , the mains potential of 110V will appear as a 10mV noise input. This value is well in excess of the ECG signal itself.

The key to extracting the desired ECG signal from the 60Hz noise is the fact that the ECG signal is the difference in potential between a pair of electrodes. Rejection of main interference therefore depends on the use of a differential amplifier in the input stage of the ECG machine, the amount of rejection depending on the ability of the amplifier to reject common-mode voltages [44]. For this reason INA128 has been selected. The INA128 is low power, general purpose instrumentation amplifiers offering excellent accuracy. The versatile 3-op amp design and small size make them ideal for a wide range of applications. Current-feedback input circuitry provides wide bandwidth even at high gain (200kHz at $G = 100$). A single external

resistor sets any gain from 1 to 10,000. The INA128 provides an industry-standard gain equation

The INA128 is laser trimmed for very low offset voltage (50uV), drift (0.5uV/C) and high common-mode rejection (120dB at $G \geq 100$). It operates with power supplies as low as 2.25V, and quiescent current is only 700uA ideal for battery-operated systems. Internal input protection can withstand up to 40V without damage [45]. In Fig 39. internal design and gain calculation of INA128 is illustrated.

As it is illustrated in Fig 40. Output of Each Sensors connected with common mode and differential mode filtering to the inputs of the INA128. R2,C2 for first sensor and R3,C3 for second sensor are establishing common mode rejection. The cutoff frequencies for the filters are shown in equations below.

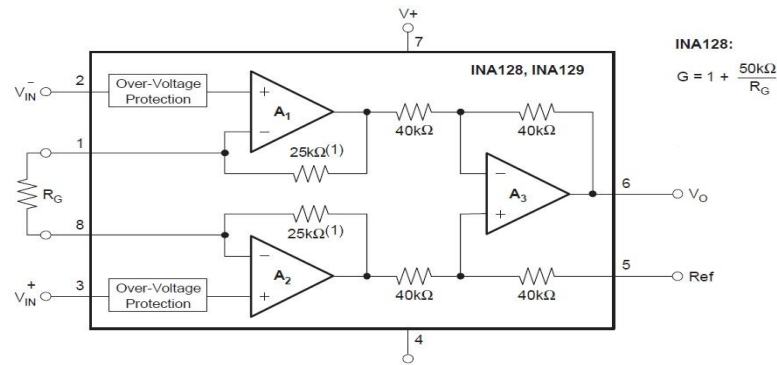


FIGURE 39. Internal Design on INA128.

For best Performance, C1 approximately ten times larger than C2 and C3.

These calculations presume $R2 = R3$ and $C2 = C3$ [43].

$$\text{Common mode cutoff Frequency} = \frac{1}{2\pi \times R2 \times C2} \quad (\text{Equation 5})$$

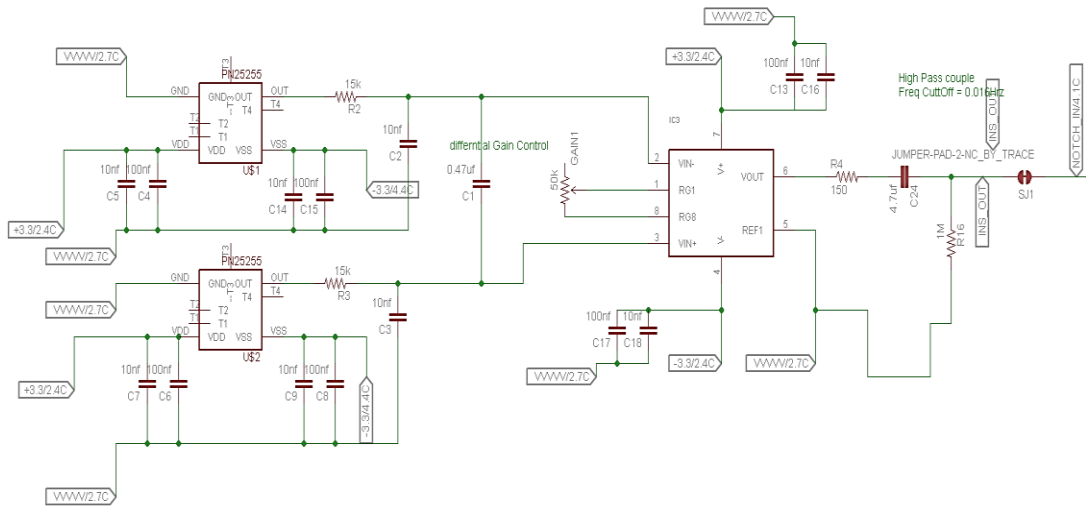


FIGURE 40. Instrumentation amplifier block schematic.

Which in this design with $R2 = 15K$ and $C2 = 10nf$ cutoff frequency is 1061.12 hertz. This cutoff frequency limits the bandwidth in the very beginning of the amplification and makes further signal conditioning much easier and much efficient [43].

$$\text{Differential mode cutoff frequency} = \frac{1}{2\pi(R2+R3)(C1+\frac{C2}{2})} \quad (\text{Equation 6})$$

As it is shown in schematic at Fig 40, $C1$ value is 0.47uf which means differential mode cutoff Frequency is 11.28 hertz which means any errors bigger than this frequency after common mode rejection filter will be eliminated here.

A 50k Ω potentiometer is used for Gain1 control stage in instrumentation amplifier instead of fix resistor. It helps to control the V_{p-p} of the output and calibrate the signal level.

While most of the interference of electrical power system is rejected by the instrumentation common mode profile but still it is visible at the output of the

instrumentation amplifier. Without reducing this 60 hertz noises to minimum it will amplify at other stages and will be visible at the output of the ECG front end.

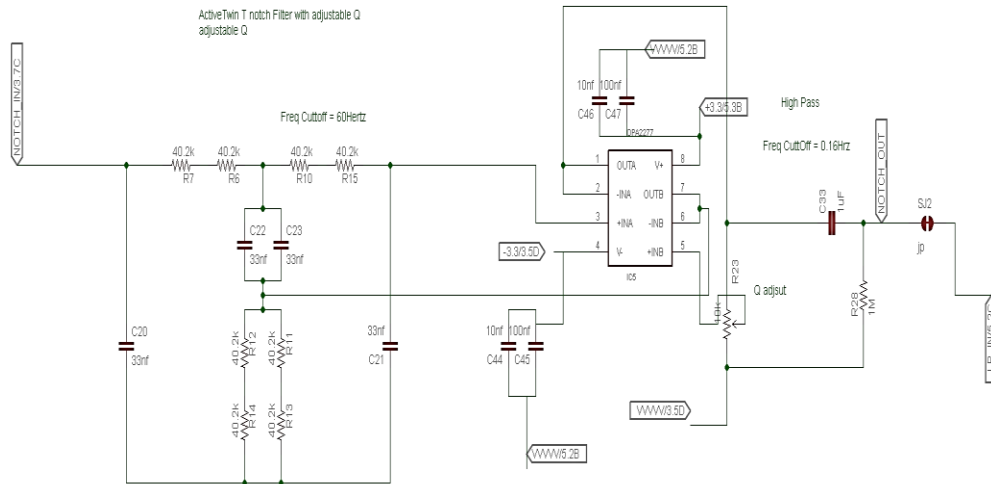


FIGURE 41. Notch (band-Stop) filter design.

To solve this problem an active Twin T Notch (band-stop) with variable Quality Factor filter placed after the instrumentation amplifier which illustrated in Fig 41. The simplified circuit of Fig. 41 is Fig. 42.

For this purpose a dual op-amp OPA2277 has been selected in regard to its ultra-low power offset voltage which is less than 10uV, very low drift $\pm 0.1\mu\text{V}/^\circ\text{C}$, Low power supply from $\pm 2\text{V}$ and low power consumption characteristics [47].

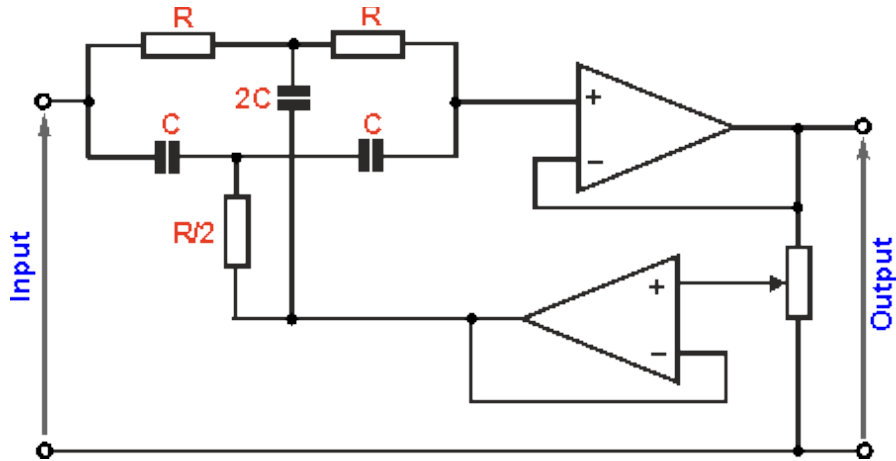


FIGURE 42. Notch filter equivalent circuit.

While all resistors and capacitors are high quality and they just have 0.05% tolerance, even these small errors can cause cutoff frequency drift from 60hertz. A variable quality factor with “Q adjust” potentiometer gives flexibility to the board to minimize the errors and improves the efficiency of the filter.

The cutoff frequency of Twin T notch filter can be calculated by the equation 7 [46]:

$$F_{notch} = \frac{1}{2\pi RC} \quad (\text{Equation 7})$$

In this design $R = 80.4k \Omega$ and $C = 33nf$ and so, $F_{notch} = 59.98603$ hertz .

6 order Sallen key order active low pass filter. The useful bandwidth of an ECG signal, depending on the application, can range from 0.5 Hz to 50 Hz for a monitoring application in intensive care units—up to 1 kHz for late-potential measurements (pacemaker detection). A standard clinical ECG application has a bandwidth of 0.05 Hz to 100 Hz. To guarantee full acceptable bandwidth the low pass filter cutoff frequency is 140 Hz [48].

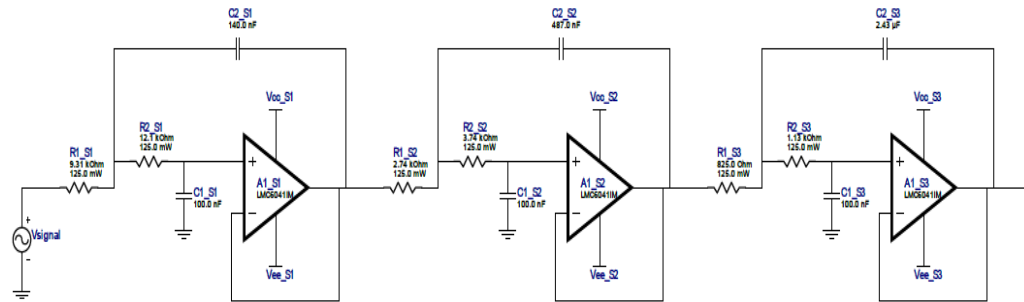


FIGURE 43. Sallen_key 6 Order low pass filter schematic.

To achieve acceptable signal quality 6 order Sallen_key low pass filter with -60db attenuation has been designed [49]. For this purpose which is illustrated in Fig 43. LMC6464 chip from Texas Instruments has been selected.

LMC6464 is a micro power version of the popular LMC6482/4, combining Rail-to-Rail Input and Output Range with very low power consumption. The LMC6462/4 provides an input common-mode voltage range that exceeds both rails. The rail-to-rail output swing of the amplifier, ensured for loads down to 25 kΩ, assures maximum dynamic signal range. This rail-to-rail performance of the amplifier, combined with its high voltage gain makes it unique among rail-to-rail amplifiers. The LMC6462/4 is an excellent upgrade for circuits using limited common mode range amplifiers. The LMC6462/4, with ensured specifications at 3V APPLICATIONS and 5V, is especially well-suited for low voltage applications. A quiescent power consumption of 60 μW per amplifier (at VS = 3V) can extend the useful life of battery operated systems. The amplifier's 150fA input current, low offset voltage of 0.25 mV, and 85 dB CMRR maintain accuracy in battery-powered systems.

The 4th available op-amp in the LMC6464 has been used for final stage gain and also offset control of the output Signal [49].

As shown in Fig 44. There is two 50k Ω potentiometers for signal offset control and the gain control. These two variables, gains and offset is useful to shift the negative part of the ECG signal to positive side and keep the signal between 0volt to 3.3volts to be able to sample full pattern with the ADC.

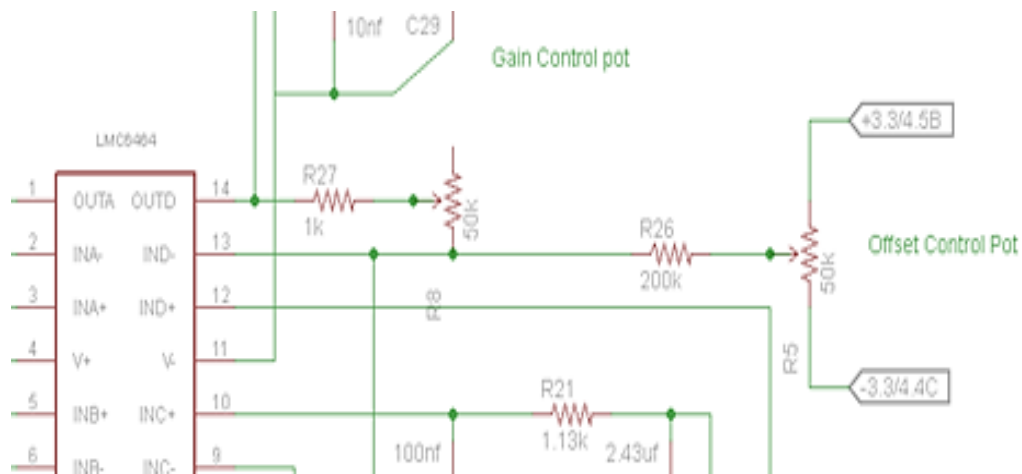


FIGURE 44. Gain and offset control.

CHAPTER 4

RESULTS

Data Acquisition

For illustrating and analyzing ECG signal, a test firmware has been developed. This code samples the analog output signal of ECG board with LPC 1549's internal 12 bit analog to digital input as illustrated in Figure 45.

```
int avg;
Timeout ecgTimer;
int16_t voltage ,i;

void plotECG()
{
    ecgTimer.detach(); //diabile ECG timerout interrupt
    tmp = 0;
    for (int i = 0 ; i < 10 ; i++) //avvrage 10 fast sampleing
    {
        //      tmp +=(ecg.read());
        //      wait_us(100);
    }
    avg = ((tmp*3000/10)); // multiple by reference voltage 3 volts
    and sclae up by 1000 times
    //      printf("%d\r\n",avg); // Send To PC with Bluetooth
    ecgTimer.attach_us(&plotECG,700); // Start and wait for timerout
    again
}

int32_t main() {
    rn42RST = true; //Turn on RN42 Bluetooth
    serial.baud(115200); // set the baud rate
    wait_ms(40); // settle down

    serial.printf("hello\r\n"); //Say Hello To everybody
    ecgTimer.attach_us(&plotECG,700); //Start sampleing with 1khz
    sample rate
    while(1)
    {
        /// nothing just timer interrupts in background
    }
}
```

FIGURE 45. ECG sampling test code.

Measured Sampling rate fixed to 1 khz which each sample is the average of 10 fast sample to reduce potential high frequency noises on ADC input. Sampling data

streams from ECG have been collected by PC data logger application and then it has been stored in file for further analysis. The ECG output signal for Person A is illustrated in Fig 46. The signal shown In Fig 46. Depicts raw ECG signal without any digital signal processing and contains common ECG signal issues such as power line interference, baseline wander [50], and high frequency noises.

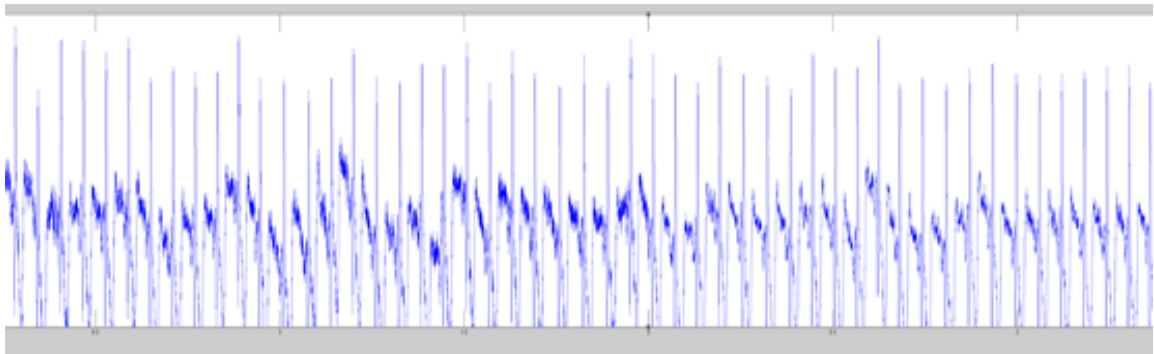


FIGURE 46. Raw collected ECG signal.

Collected Data Analysis

To remove these noises and to get proper ECG signal, we have used Pan Tompkins QRS detection algorithm [51]. This is available as Pan Tompkins toolset in MATLAB [52].

First graph in Fig 47 depicts purified ECG signal after executing internal band-pass filter, derivative filter and baseline wander removal. By comparing the filtered signal in first graph at Fig 47 with the raw ECG signal at Figure 46, it is clearly seen the ECG signal noises and baseline instability has been removed. In the first and second graph at Fig 47, red marked dots on the signals are detected as QRS peaks. Moreover, all ECG pulse train intervals have been detected in the third graphs (of Fig 47) by red perpendicular lines. This ECG signal has been collected from

person A for more than 60 seconds. In these three graphs, none of the QRS peaks and pulse train intervals missed by the well-known Pan Tompkin algorithm which confirms that the designed portable and power efficient ECG system is capable of providing medical graded data for wearable biomedical applications development.

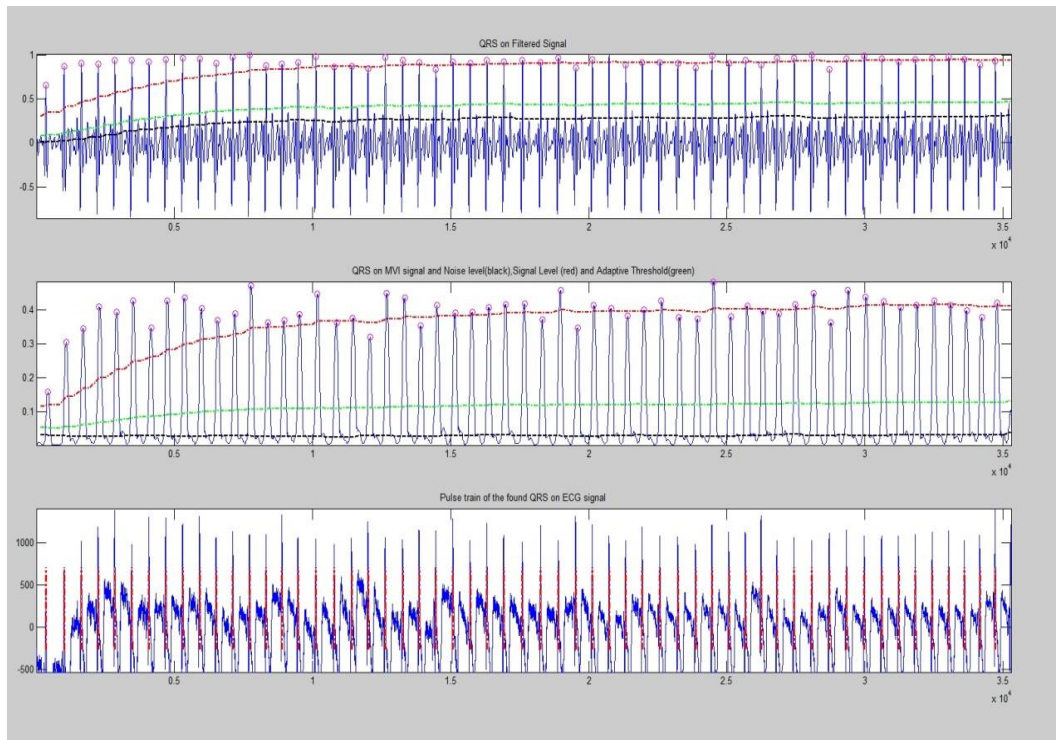


FIGURE 47. QRS detection results.

CHAPTER 5

CONCLUSION

Living in an era characterized by the rapid growth of the internet, people can now easily access information about anything at anytime and anywhere. Smart technologies are enabling people to connect phones, appliances, homes, cities, and anything else with internet access, to each other, thus enhancing our quality of life. Still, today, what is missing is that people cannot get relatively basic information about their own bodies without visiting a physician. Moreover, there is no wide-spread system to provide vital health information to individuals without visiting health care provider facilities. This thesis proposes the implementation of a device based layered body sensor framework which could be connected with Cloud infrastructure to collect and provide health data that might be a paradigm shift for smart health.

Recent BSNs have enabled to monitor the patients by gathering data streams and store them locally in the device or in a personal computer. However, there are significant challenges regarding the large amount of data and the limited processing power of the BSN. We foresee that leveraging the sensor-cloud infrastructure combined with proposed portable device would be able to tackle these challenges. Hence as a future expansion, we propose an integrated platform that will offer: (a) the ability to utilize heterogeneous sensors, (b) scalability of data storage, (c) scalability of processing power for different types of analysis, (d) global access to the processing and storage infrastructure, and (e) accessible results for BSN services. Various machine learning algorithms could be employed to analyze the stored data in the Cloud to provide valuable suggestions for better health.

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